

Recovering the original prime calculation over full Split-Zero support

Programme continuation with explicit source provenance

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This working paper reconstructs specified parts of the original arithmetic programme over its actual lattice coefficient object $G_L(\mathbb{C})$. The previously used $G_{\mathbb{B}}(\mathbb{C})$ is a Boolean special case; its spectrum is not a description of the full support spectrum. The foundational lattice results below are credited to the retained Version 11 source. The new calculations construct the exact recovered carriers, maps, original metric, mixed kernels, and the map from prime-support defects to Weil's local distribution.

The central algebraic recovery is the path closure of the original companion operator. Its support graph forces a synchronized carrier which retains the entire lattice L . The original linear maps and prime group action lift on that carrier, and all preimages of its projection from the full free semimodule are calculated. Supported zero maps remain distinct from actual absence. The calculation also exhibits a higher mixed kernel invisible to individual branch linearizations and computes an exact inverse recovering all joint support moments. An explicit compact test detects any prescribed prime-support defect, with both Weil moment conditions satisfied.

The recovery also supplies a pointed category whose zero morphisms are the retained supported zeros. Its exact kernels, cokernels and cohomology recover the original complex. For the actual companion prime matrices, an invertible lift to the entire free semimodule is impossible; a proved fibre groupoid and its prime correspondences retain every mask with their precise composition laws. The full additive support jet and all extensions of the earlier scalar action are calculated. Joint tensor observations then recover every finite mixed support coefficient and connect the support-addition operators to the diagonal trace insertions through an explicit incidence algebra.

These are recovery and defect calculations, not a completed proof or disproof of the Riemann hypothesis. No sign for the full-place Weil form is assumed. In particular a counterexample to a restricted place functional is explicitly identified as such.

Foundational credit. The scalar prime chain $(\tau) \subsetneq (e) \subsetneq (p)$, including the nonzero prime element e and $(e) = \{\tau, e\}$, is an established result of the first globalization paper. The user identifies Gemini 2.5 Pro as the originating model for that result. This attribution is recorded as user-provided provenance. The exact original `globalization_note.tex` has an empty author field; the later Version 11 source has the byline The Clankers. These source identities are retained separately. Repetition of that prime-chain theorem is not counted as a new result here. The human sources used for the analytic maps are Alain Connes and André Weil, and the lattice background includes Garrett Birkhoff. The original ES–Fable lineage remains credited to Levent Alpöge where its transport is used; this paper does not reassign that construction to its own recovery calculations.

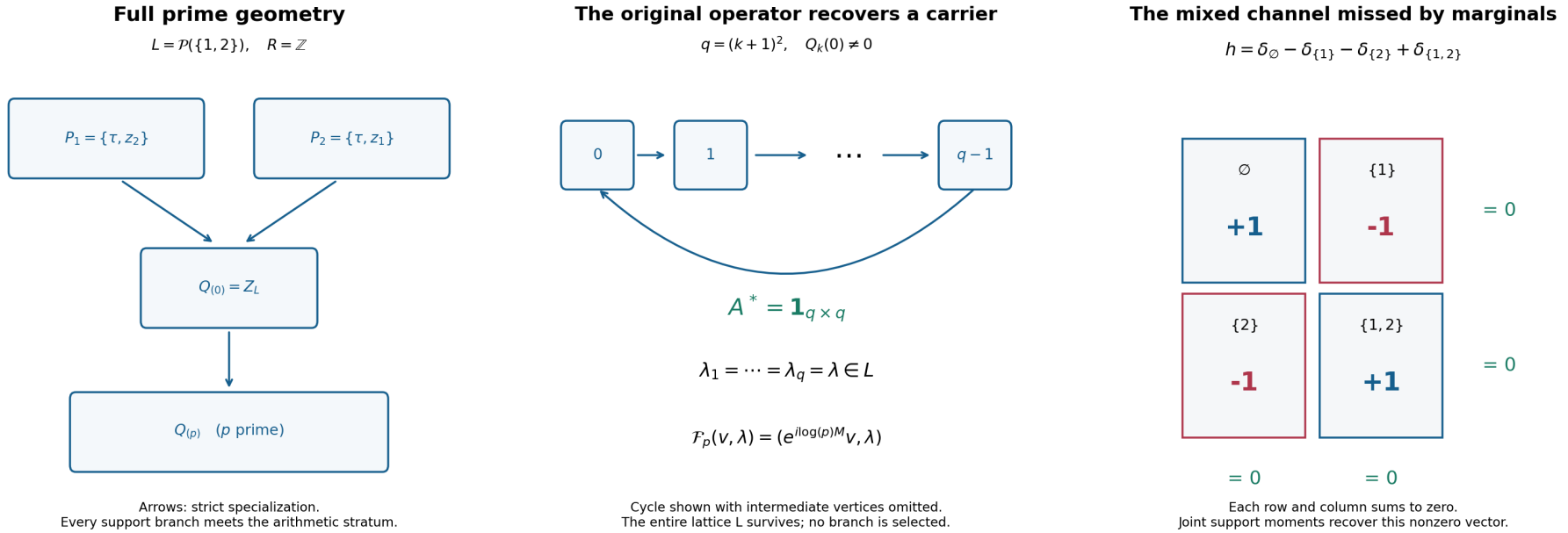
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Full support, recovered prime evolution, and the remaining mixed channel



Proofs: FLS2-3; CLM8-17; FLC6-11,20-21. Foundation: The Clankers, Split Support Geometry v11; Birkhoff lattice coordinates.

Figure 1: Left: the full two-channel prime spectrum, using the source's existing ordinal-sum theorem. Middle: the original companion cycle and its path closure, proved in CLM8–17. Right: the signed four-state vector in FLC10, whose branch marginals vanish, with full recovery in FLC11. The middle cycle abbreviates intermediate vertices; the left arithmetic node ranges over every rational prime.

1 The full support lattice: stalks, residues, and exact branch recovery

Original source and correction of scope. The source is The Clankers, *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, June 2026, pre-publication draft Version 11, file `split_support_geometry_arithmetic_curve_v11.tex`, SHA-256 5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d. Its Definition `def:lattice-split` and the existing theorems on prime classification, the spectral ordinal sum, support localization, Birkhoff support coordinates, and the branch-selected spectrum already construct and prove the full lattice model, its spectrum, finite support coordinates, and branch selection. Those are the source's results, not new discoveries of this continuation. Its bibliography credits Garrett Birkhoff, *Lattice Theory*, third edition (1967), for the lattice-theoretic background. The local calculations and explicit infinite-point distinctions below apply that source without replacing its lattice by one Boolean branch.

Let R be a nonzero commutative unital ring and L a nontrivial bounded distributive lattice. The original object is

$$A = G_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\},$$

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu). \quad (\text{FLS1})$$

Write $Z_L = \{z_\lambda : \lambda \in L\}$, $\tau_L = z_{0_L}$ and $e_L = z_{1_L}$. The source's prime and topology theorem gives

$$P_{\mathfrak{p}} = \{z_\lambda : \lambda \in \mathfrak{p}\} \quad (\mathfrak{p} \in \text{SpecLat } L), \quad Q_{\mathfrak{q}} = Z_L \cup \{(r, 1_L) : r \in \mathfrak{q}\} \quad (\mathfrak{q} \in \text{Spec } R), \quad (\text{FLS2})$$

with every $P_{\mathfrak{p}}$ strictly below every $Q_{\mathfrak{q}}$. Its principal opens are

$$D(z_\lambda) = \{P_{\mathfrak{p}} : \lambda \notin \mathfrak{p}\}, \quad D((r, 1_L)) = \text{SpecLat } L \cup D_R(r). \quad (\text{FLS3})$$

In the following, stalk means the prime localization of the usual sheaf of local fractions for this semiring-prime spectrum. None of these local computations substitutes the underlying multiplicative-monoid spectrum or the twisted prime-congruence spectrum for (FLS2).

1.1 Support stalks are localized lattices, not generally Boolean

Fix $\mathfrak{p} \in \text{SpecLat } L$ and set $F = L \setminus \mathfrak{p}$. This is the complementary prime filter. Since $e_L \notin P_{\mathfrak{p}}$, first invert e_L . The source's support-localization theorem gives $A[e_L^{-1}] \cong L$, and a supported amplitude maps to 1_L . The remaining denominators are exactly the elements of F . Hence the full support stalk is

$$A_{P_{\mathfrak{p}}} \cong L[F^{-1}]. \quad (\text{FLS4})$$

This can be written without unspecified fractions. An idempotent which becomes invertible becomes the unit, so every element of F becomes 1. Every fraction is therefore represented by an element of L . The exact equality relation is

$$\lambda \sim_F \mu \iff \exists \nu \in F : \nu \wedge \lambda = \nu \wedge \mu, \quad L[F^{-1}] = L / \sim_F. \quad (\text{FLS5})$$

The equivalence follows directly from multiplicative fraction equality. It is also a lattice congruence: reflexivity uses $1_L \in F$; symmetry is immediate; transitivity uses the meet of two witnesses, which remains in F . Meeting a witness with a common factor proves compatibility with meets, and distributivity proves compatibility with joins. All joins, meets, zero and unit in this stalk are therefore explicit.

The elements outside $\mathfrak{p}$ all have class 1. An element $\lambda \in \mathfrak{p}$ cannot have class 1: a witness would give $\nu \wedge \lambda = \nu$ with $\nu \in F$, hence $\nu \leq \lambda$ and $\nu \in \mathfrak{p}$, a contradiction. Thus the stalk has the unique maximal ideal

$$\mathfrak{m}_{\mathfrak{p}} = \{[\lambda]_F : \lambda \in \mathfrak{p}\}, \quad L[F^{-1}] /_{\text{Bourne}} \mathfrak{m}_{\mathfrak{p}} \cong \mathbb{B}. \quad (\text{FLS6})$$

For maximality, the only multiplicative unit of any lattice semiring is its top: an invertible idempotent is the unit. All other classes here lie in $\mathfrak{m}_{\mathfrak{p}}$, which is an ideal. For the quotient statement, the Bourne congruence identifies every element of this ideal with zero, while all elements outside it are already 1. It cannot identify 1 with zero: an equality $1 \vee i = j$ with both $i, j \in \mathfrak{m}_{\mathfrak{p}}$ would put 1 in the ideal. This proves the two-element residue. It does not prove that the stalk before quotienting has two elements.

For finite L , the source's finite Birkhoff theorem identifies F with the principal filter $\uparrow j$ of a unique nonzero join-irreducible element j . Then

$$L[F^{-1}] \xrightarrow{\cong} \downarrow j, \quad [\lambda]_F \mapsto \lambda \wedge j, \quad \downarrow j = \{\mu : \mu \leq j\}, \quad (\text{FLS7})$$

where the unit on the right is j . A witness in (FLS5) implies equality after meeting with j , since j is below every witness. Conversely equality after meeting with j uses j itself as witness. Surjectivity follows by taking $\lambda = \mu \leq j$. Distributivity proves that this bijection preserves both operations. Thus it is the claimed semiring isomorphism, with every support below j retained.

For the source's chain example $L = \{0 < a < 1\}$ this gives

$$A_{P_{\{0\}}} \cong \{0, a\} \cong \mathbb{B}, \quad A_{P_{\{0, a\}}} \cong \{0, a, 1\}. \quad (\text{FLS8})$$

The second stalk is not Boolean, although its residue (FLS6) is Boolean. This is an explicit counterexample to the assertion that every support stalk is $\mathbb{B}$.

If L is any Boolean algebra, finite or infinite, every support stalk is instead $\mathbb{B}$. Indeed, for $\lambda \in \mathfrak{p}$, its Boolean complement $\neg \lambda$ is in F , and $(\neg \lambda) \wedge \lambda = 0$. Formula (FLS5) therefore makes $[\lambda]_F = 0$. Elements in F have class 1, so the exact isomorphism is the branch itself:

$$L[F^{-1}] \cong \mathbb{B}, \quad [\lambda]_F \mapsto \epsilon_{\mathfrak{p}}(\lambda) = \mathbf{1}_{\lambda \notin \mathfrak{p}} \quad (L \text{ Boolean}). \quad (\text{FLS9})$$

1.2 Arithmetic stalks retain the entire lattice

Fix $\mathfrak{q} \in \text{Spec } R$. The complement of $Q_{\mathfrak{q}}$ consists precisely of the supported amplitudes $(s, 1_L)$ with $s \notin \mathfrak{q}$. The complete localization is

$$A_{Q_{\mathfrak{q}}} \xrightarrow{\cong} G_L(R_{\mathfrak{q}}), \quad \frac{(r, 1_L)}{(s, 1_L)} \mapsto (r/s, 1_L), \quad \frac{z_{\lambda}}{(s, 1_L)} \mapsto z_{\lambda}. \quad (\text{FLS10})$$

The fraction operations verify the map. Equality between supported fractions is exactly ring fraction equality; when their amplitude becomes zero the image is z_{1_L} . Multiplication by any supported denominator leaves each z_{λ} unchanged, so two such lower-support fractions are equal precisely when their lattice labels are equal. A supported fraction cannot become z_{λ} for $\lambda < 1_L$, because every fraction-equality witness preserves the support 1_L on its supported side. These facts prove injectivity; all displayed target elements have the given preimages, proving surjectivity.

The supported and amplitude residue maps consequently are

$$G_L(R_{\mathfrak{q}}) \rightarrow G_L(\kappa(\mathfrak{q})) \xrightarrow{p_L} \kappa(\mathfrak{q}). \quad (\text{FLS11})$$

The first identifies represented amplitudes with the same residue but keeps all z_{λ} distinct. The composite is the Bourne quotient by the maximal amplitude ideal, which identifies the entire zero lattice with the field zero. This proves the different residue fibres exactly. For $R = \mathbb{Z}$, both the generic arithmetic stalk $G_L(\mathbb{Q})$ and every finite-place stalk $G_L(\mathbb{Z}_{(p)})$ still contain the full lattice L .

1.3 A single branch and all branches have different exact fibres

Let $X = \text{SpecLat } L$ and write $\epsilon_x : L \rightarrow \mathbb{B}$ for its branch at x . The source's branch quotient is $\beta_x : G_L(R) \rightarrow G_{\mathbb{B}}(R)$, acting as identity on the represented amplitudes and as ϵ_x on the zero

lattice. Its spectral inclusion selects exactly $\{x\} \blacktriangleright \text{Spec } R$. The maps on its stalks, calculated above, are

$$\begin{aligned} L[(L \setminus x)^{-1}] &\rightarrow \mathbb{B} \quad \text{at the selected support point,} \\ G_L(R_q) &\rightarrow G_{\mathbb{B}}(R_q) \quad \text{at an arithmetic point.} \end{aligned} \quad (\text{FLS12})$$

The first is the residue map (FLS6); the second applies ϵ_x to the entire remaining zero lattice. These are the exact maps by which choosing one branch can lose support even when the amplitude is unchanged.

For finite L , the source already proves that all branches recover L : their coordinate patterns are the order ideals of its join-irreducible poset $J(L)$. We retain that result and extend its concrete comparison to arbitrary bounded distributive L . The needed prime-separation argument is included. If $\lambda \not\leq \mu$, the lattice ideal $\downarrow \mu$ and filter $\uparrow \lambda$ are disjoint. By the maximal principle choose an ideal I maximal among those containing the first and avoiding the second. It is prime. Indeed if $a, b \notin I$, the ideals obtained by adjoining a and b each meet the filter. There are f, g in the filter and $i, j \in I$ such that $f \leq i \vee a$ and $g \leq j \vee b$. Distributivity gives

$$f \wedge g \leq (i \vee a) \wedge (j \vee b) \leq i \vee j \vee (a \wedge b). \quad (\text{FLS13})$$

If $a \wedge b \in I$, the right-hand side belongs to I , so the filter would meet I , a contradiction. Thus I is prime, contains μ , and omits λ . This is the classical distributive-lattice prime-separation argument, used here with its choice principle explicit.

It follows that the simultaneous branch map is injective:

$$\beta : G_L(R) \hookrightarrow \prod_{x \in X} G_{\mathbb{B}}(R), \quad a \mapsto (\beta_x(a))_{x \in X}. \quad (\text{FLS14})$$

Distinct nonzero amplitudes, or a nonzero and a zero amplitude, are separated by any coordinate. Two elements of the zero lattice are separated by (FLS13). Its image is exactly the families with a common amplitude r such that, when $r = 0$, the set of coordinates having supported zero is $D_L(\lambda)$ for one $\lambda \in L$:

$$\text{im } \beta = \{((r, 1))_{x \in X} : r \in R\} \cup \{((0, \epsilon_x(\lambda)))_{x \in X} : \lambda \in L\}. \quad (\text{FLS15})$$

This gives the exact consistency restriction on a family of branches; arbitrary independent branch values are not automatically a full-support element.

The allowable support patterns in (FLS15) are exactly the compact opens of X . The identities $D_L(\lambda \vee \mu) = D_L(\lambda) \cup D_L(\mu)$ and $D_L(\lambda \wedge \mu) = D_L(\lambda) \cap D_L(\mu)$ follow from the prime-ideal axioms. To prove compactness, suppose $D_L(\lambda) \subseteq \bigcup_i D_L(\mu_i)$ has no finite subcover. Then λ is not below any finite join of the μ_i , by the separation argument. The ideal generated by all the μ_i is disjoint from $\uparrow \lambda$; applying the same maximal-ideal proof produces a prime containing every μ_i and omitting λ , contradicting the cover. Every compact open has a finite cover by basic opens, so it is itself one basic open. We have proved the exact recovery map

$$L \xrightarrow{\cong} \mathcal{KO}(X), \quad \lambda \mapsto D_L(\lambda), \quad (\text{FLS16})$$

where $\mathcal{KO}(X)$ is the lattice of compact open subsets with union and intersection. This is the classical spectral representation applied to the original zero lattice, not a replacement of that lattice by a single stalk. A chosen branch alone records only whether $x \in D_L(\lambda)$, identifying all labels on each of the two sides of that membership test.

For finite L the lost-support objects can be given still more explicitly using the source's Birkhoff coordinates. Let $J = J(L)$, retain only a subset $T \subseteq J$ of the branches, and suppose the observed positive coordinates are $U \subseteq T$. The entire zero-support fibre is

$$\{I \in \text{Idl}(J) : I \cap T = U\} = [\downarrow U, J \setminus \uparrow (T \setminus U)] \subseteq \text{Idl}(J), \quad (\text{FLS23})$$

where the interval is empty when its lower bound is not contained in its upper bound. To prove this, containing U forces the order ideal I to contain $\downarrow U$. Omitting every point of $T \setminus U$ forces it to omit their upward closure. Conversely every order ideal between these two bounds contains the required positive coordinates and none of the negative coordinates. The lower and upper bounds are themselves order ideals, so this is the exact interval of omitted information, with its inherited joins and meets and its inclusion into L through Birkhoff's map. For a single branch j the two fibres are therefore

$$\epsilon_j^{-1}(0) \cong [\emptyset, J \setminus \uparrow j], \quad \epsilon_j^{-1}(1) \cong [\downarrow j, J]. \quad (\text{FLS24})$$

Every branch is necessary for full reconstruction when L is finite: if j is omitted, the two order ideals $\downarrow j$ and $(\downarrow j) \setminus \{j\}$ have identical values at every retained branch and are distinct. This proves the failure of that observation by constructing its exact fibre, rather than replacing the lost states by an unspecified error.

1.4 Infinite Boolean branches and complete-frame points

For $L = \mathcal{P}(I)$, bounded lattice homomorphisms $L \rightarrow \mathbb{B}$ are exactly ultrafilters U on I , with $\epsilon_U(A) = \mathbf{1}_{A \in U}$. This follows because finite meets and joins give a prime filter, and complementation gives exactly one of $A, I \setminus A$ in it. Conversely that ultrafilter rule preserves the two finite operations. All these branches belong to the semiring-prime spectrum used in (FLS2), including nonprincipal ultrafilters when they exist. A point of the *complete frame*, however, must preserve arbitrary joins. Such a branch is exactly evaluation at a single $i \in I$:

$$\epsilon_i(A) = \mathbf{1}_{i \in A}, \quad \epsilon_U \left(\bigcup_{i \in I} \{i\} \right) = 1 \neq \bigvee_{i \in I} \epsilon_U(\{i\}) = 0 \quad (U \text{ nonprincipal}). \quad (\text{FLS17})$$

The displayed equation proves failure of complete-join preservation for a nonprincipal branch. Conversely a complete-join preserving branch applied to $I = \bigcup_i \{i\}$ must send some singleton to 1; the filter laws then make it evaluation at that element. Evaluation branches alone already recover $\mathcal{P}(I)$ by all its membership coordinates, but they are not all the bounded-lattice prime points when I is infinite.

More generally, for a complete Boolean algebra L , complete-frame points are exactly its atoms. A bounded lattice branch preserves complements. If it also preserves arbitrary joins, De Morgan's law shows it preserves arbitrary meets. For its filter F , put $a = \bigwedge_{f \in F} f$. Its value at a is 1, so $a \neq 0$ and $a \in F$. If $0 < b < a$, either $b \in F$, contradicting minimality of a , or $\neg b \in F$, contradicting $a \leq \neg b$. Thus a is an atom and the branch is $\lambda \mapsto \mathbf{1}_{a \leq \lambda}$. Conversely an atom supplies a completely join-preserving branch: finite meet with that atom distributes over any join, and each resulting meet is either zero or the atom. Therefore

$$\text{Hom}_{\text{Frm}}(L, \mathbb{B}) \cong \text{Atoms}(L), \quad \text{Hom}_{\text{DLat}}(L, \mathbb{B}) \cong \text{Ultrafilters}(L) \quad (L \text{ complete Boolean}). \quad (\text{FLS18})$$

An atomless nontrivial complete Boolean algebra consequently has no complete-frame points, while its bounded-lattice branches still separate elements by (FLS13). Empty frame point data cannot reconstruct that algebra. This is a change of the permitted morphisms, not disappearance of the original lattice or its semiring-prime spectrum.

1.5 Exact corrections to the Boolean-only calculation

The earlier TPW calculation concerns $G_{\mathbb{B}}(R)$. It remains a calculation of one Boolean quotient, and must be so labelled. In the full source,

$$G_L(0) = L, \quad G_L(R)[e_L^{-1}] = L, \quad Q_{(0)} = Z_L \quad (R \text{ a domain}). \quad (\text{FLS19})$$

The first is immediate from (FLS1), the second is the source's localization theorem, and the third is (FLS2). In particular the arithmetic zero prime contains every mixed support label, not merely τ_L and e_L . Moreover absence by itself need not be prime. For $L = \mathcal{P}(\{1, 2\})$,

$$z_{\{1\}}z_{\{2\}} = \tau_L, \quad z_{\{1\}} \neq \tau_L \neq z_{\{2\}}, \quad (\text{FLS20})$$

so $\{\tau_L\}$ is not prime. The support stratum has both coordinate branches, with neither a substitute for the other. The unchanged arithmetic residue amplitude alone therefore does not describe the changed source spectrum or prove a statement about every original mixed-support operator.

The compact-carrier test algebra constructed in TPW is a different specified support object: its carrier multiplication is set product on the multiplicative test line. The original zero lattice has meet as its multiplication. Their exact respective products are

$$e_S * e_T = e_{ST} \quad \text{in the TPW carrier algebra,} \quad z_\lambda z_\mu = z_{\lambda \wedge \mu} \quad \text{in } G_L(R). \quad (\text{FLS21})$$

TPW supplies a map to the source's single-fibre Boolean model, not an identification with the entire lattice model. Full original-lattice convolution or trace claims must retain L and calculate their branch maps, including which finite or infinite joins those maps preserve. In particular finite sums are always detected by all branches, whereas a chosen infinite-join observation cannot be moved through a nonprincipal branch without checking it; (FLS17) is an explicit failure calculation.

The surviving exact comparison, both before and after arithmetic localization, is

$$\begin{array}{ccc} G_L(R) & \hookrightarrow & \prod_{x \in X} G_{\mathbb{B}}(R) \\ \downarrow & & \downarrow \\ G_L(R_q) & \hookrightarrow & \prod_{x \in X} G_{\mathbb{B}}(R_q). \end{array} \quad (\text{FLS22})$$

Every map is the explicit amplitude localization or branch map proved above, so the square commutes on $(r, 1_L)$ and every z_λ , hence on all elements. Injectivity of each horizontal map follows from prime separation. This retains the full lattice through all its compatible branches and specifies exactly the data lost when only one branch is used.

2 Exact recovery of the amplitude constructions over the full lattice

Source and scope. The coefficient object in this section is the actual $G_L(\mathbb{C})$ of The Clankers, *Split Support Geometry*, Version 11, Definition `def:lattice-split`, lines 1226–1249 of the retained original TeX. The complete source is retained as `sources/FULL_LATTICE_FOUNDATION_V11.tex`. Its existing prime, localization and branch theorems are antecedents, not results first discovered in this continuation. We now construct the maps that recover the existing amplitude linear algebra over this coefficient object. This construction gives a specified subobject of the full free semimodule; it does not identify those two objects.

Let L be a nontrivial bounded distributive lattice and put $S_L = G_L(\mathbb{C})$. Write $z_\lambda = (0, \lambda)$, $\tau_L = z_0$ and $e_L = z_1$. For a complex vector space V define

$$V_L^\uparrow = \{(0_V, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\} \subset V \times L. \quad (\text{FLR1})$$

Its operations, with every zero retained, are

$$(v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu), \quad (c, \alpha)(v, \lambda) = (cv, \alpha \wedge \lambda). \quad (\text{FLR2})$$

These are closed operations: a nonzero sum has a nonzero summand and therefore top support; a nonzero scalar product has both factors of top support. Associativity, units and the two

distributive laws follow from the vector-space laws and distributivity in L . The actual zero is $(0_V, 0_L)$. Thus FLR1 is an S_L -semimodule. Its amplitude projection $a_V(v, \lambda) = v$ is equivariant for the ring reflection $p_L : S_L \rightarrow \mathbb{C}$.

For each original linear map $T : V \rightarrow W$, define

$$T_L^\uparrow(v, \lambda) = (Tv, \lambda). \quad (\text{FLR3})$$

When $Tv = 0$, the output keeps its original support. The formulas FLR2 prove additivity and S_L -linearity of FLR3. They also prove

$$(UT)_L^\uparrow = U_L^\uparrow T_L^\uparrow, \quad (T + T')_L^\uparrow = T_L^\uparrow + T'_L^\uparrow, \quad (\text{id}_V)_L^\uparrow = \text{id}_{V_L^\uparrow}. \quad (\text{FLR4})$$

For the sum, both output supports are λ , so their join is λ . The lifted zero linear map is

$$\varepsilon_V = (0_{V,V})_L^\uparrow, \quad \varepsilon_V(v, \lambda) = (0_V, \lambda) = e_L(v, \lambda). \quad (\text{FLR5})$$

It is distinct from the categorical zero map, which has constant value $(0_V, 0_L)$. The lift is faithful, since evaluation on $(v, 1_L)$ recovers Tv . Thus FLR4 proves actual composition identities without silently erasing the distinction in FLR5.

The lift is also compatible with every Boolean branch $\epsilon : L \rightarrow \mathbb{B}$. The map

$$b_{\epsilon,V} : V_L^\uparrow \longrightarrow V_{\mathbb{B}}^\uparrow, \quad (v, \lambda) \longmapsto (v, \epsilon(\lambda)) \quad (\text{FLR6})$$

is equivariant for the source's coefficient map β_ϵ , and $b_{\epsilon,W} T_L^\uparrow = T_{\mathbb{B}}^\uparrow b_{\epsilon,V}$. This follows by substituting FLR2 and using preservation of join and meet. It specifies exactly how every branch calculation receives the full-lattice calculation; one such square does not recover all of L .

2.1 The original metric, section and full supported kernel

Take an original Hermitian vector space (V, h_V) , with the convention that the first variable is conjugate-linear. Its lifted pairing is

$$H_{V,L}((v, \lambda), (w, \mu)) = (h_V(v, w), \lambda \wedge \mu) \in S_L. \quad (\text{FLR7})$$

If its amplitude is nonzero, both vectors have top support, so the displayed pair lies in S_L . The Hermitian symmetry, additivity and sesquilinearity follow respectively from those of h_V , distributivity in L , and FLR2. In particular

$$H_{W,L}(T_L^\uparrow x, y) = H_{V,L}(x, (T^\dagger)_L^\uparrow y). \quad (\text{FLR8})$$

Both amplitudes are the original adjoint identity and both supports are the same meet. At a zero-amplitude vector $(0, \lambda)$ its self-pairing is z_λ , not necessarily τ_L . No order on S_L is being invented: all numerical metric inequalities here refer to the exact original real amplitudes under p_L .

Apply FLR1–8 at every original cutoff to the spaces E, B, K , the map $\Lambda : E \rightarrow B$, the complete metric G_N , and the maps of ST2. To avoid confusing the source's section with the support lattice, write L_{sec} for precisely its original $G_N^{-1} \Lambda^* Q_B$, with no alteration to this matrix. Then

$$\Lambda_L^\uparrow (L_{\text{sec}})_L^\uparrow = \text{id}_{B_L^\uparrow}, \quad (J^\dagger)_L^\uparrow J_L^\uparrow = \text{id}_{B_L^\uparrow}, \quad (P_B)_L^\uparrow = J_L^\uparrow (J^\dagger)_L^\uparrow. \quad (\text{FLR9})$$

These follow from the original matrix identities by FLR4. The support map $\sigma_V(v, \lambda) = \lambda$ is retained. Define $Z_L(V) = \{(0, \lambda) : \lambda \in L\}$. The full supported kernel is the exact inverse image

$$(\Lambda_L^\uparrow)^{-1}(Z_L(B)) = K_L^\uparrow = \{(k, 1_L) : k \in \ker \Lambda\} \cup Z_L(E). \quad (\text{FLR10})$$

Indeed membership is exactly $\Lambda v = 0$, with no condition deleting λ . In contrast, the inverse image of the single actual zero $(0, 0_L)$ is only $(0, 0_L)$. Using that single-point kernel would lose all nonzero vectors of K and every nonbottom support mask. FLR10 identifies the precise kernel object used in recovery.

There is a canonical isomorphism over L

$$(B \oplus K)_L^\uparrow \simeq B_L^\uparrow \times_L K_L^\uparrow, \quad ((b, k), \lambda) \longleftrightarrow ((b, \lambda), (k, \lambda)). \quad (\text{FLR11})$$

Here the fibre product means equality of support labels. If $\lambda < 1$ both amplitudes are zero; at 1 both are unrestricted. This proves bijectivity, and FLR2 proves S_L -linearity. Let ι_K be the original isometric inclusion. The original orthogonal decomposition and its inverse therefore lift to the explicit inverse maps

$$\begin{aligned} W_L((b, \lambda), (k, \lambda)) &= (Jb + \iota_K k, \lambda), \\ W_L^{-1}(x, \lambda) &= ((J^\dagger x, \lambda), (\iota_K^\dagger x, \lambda)). \end{aligned} \quad (\text{FLR12})$$

Their composites are identities because the original orthogonal decomposition is an identity and each composite preserves λ . The sum of the two pairings on the right of FLR11 has support $\lambda \wedge \mu$ in both summands; their join is the same support. Thus FLR12 preserves exactly the original metric lifted by FLR7.

In particular, the recovered source identities include

$$(\Pi_L^\uparrow)^2 = \Pi_L^\uparrow, \quad (R_L^\uparrow)^2 = \varepsilon_E, \quad (P_B)_L^\uparrow (I - P_B)_L^\uparrow = \varepsilon_E, \quad (\text{FLR13})$$

where the second equality retains the original $R^2 = 0$ as a supported zero map. It does not assert multiplication to actual absence. For $\Phi(X) = J^\dagger X J$, its full recovered composition law is

$$\begin{aligned} (\Phi(XY))_L^\uparrow &= (\Phi(X))_L^\uparrow (\Phi(Y))_L^\uparrow \\ &\quad + (J^\dagger X (I - P_B) Y J)_L^\uparrow. \end{aligned} \quad (\text{FLR14})$$

Substitution of FLR4 reduces its amplitude to insertion of $I = P_B + (I - P_B)$ and leaves the same support label on both sides. With $X = Y = \Pi$, it gives the complete original identity $A = A^2 + D^\dagger D$, lifted over L , with $D = (I - P_B)\Pi J$. The original squared norm and every original constant are unchanged under the proved amplitude map; no new bound is assumed.

2.2 What this recovery retains, and the larger carrier

Fix an original basis $V \simeq \mathbb{C}^n$, $n \geq 1$. There is an injective S_L -linear map

$$\iota_V : V_L^\uparrow \hookrightarrow S_L^n, \quad (v, 1_L) \mapsto ((v_i, 1_L))_i, \quad (0, \lambda) \mapsto ((0, \lambda))_i. \quad (\text{FLR15})$$

It is well defined at $(0, 1_L)$, and coordinatewise operations give linearity. Coordinates recover v and the common support, proving injectivity. For $n \geq 2$ it need not be surjective. For instance, $((1, 1_L), (0, 0_L))$ is a valid free vector and is not in its image. More generally the exact fibre of the amplitude projection of S_L^n over $v = (v_i)$ is

$$\prod_{i:v_i=0} L, \quad \text{with support } 1_L \text{ forced at each } v_i \neq 0. \quad (\text{FLR16})$$

There are no omitted choices: at a zero coordinate every lattice label is allowed; at a nonzero coordinate the definition of G_L forces top. For finite L the fibre has $|L|^{\#\{i:v_i=0\}}$ elements. Thus FLR15 recovers the original amplitude construction faithfully, while FLR16 describes the additional mixed directions that the full matrix construction must retain. The next section constructs the exact prime action on the synchronization retract and retains every additional free mask in the explicitly calculated fibres of that retraction.

2.3 Cohomology of the recovered support diagram

For clarity, consider an original complex (V^n, d^n) of complex vector spaces. At each support λ define a complex with terms V^n for $\lambda = 1_L$ and the zero vector space for $\lambda < 1_L$. For $\lambda \leq \mu$, use the identity when the labels agree and the unique zero map from a lower zero fibre to the top fibre otherwise. The maps commute with each d^n . Taking kernels, images and quotients at each label gives the exact support diagram

$$\mathcal{H}_\lambda^n = \begin{cases} \ker d^n / \operatorname{im} d^{n-1}, & \lambda = 1_L, \\ 0, & \lambda < 1_L. \end{cases} \quad \bigsqcup_{\lambda \in L} \mathcal{H}_\lambda^n = (H^n(V^\bullet))_L^\uparrow. \quad (\text{FLR17})$$

This is a calculation of a specified diagram, with its induced S_L -action from FLR2. The maps on its total semimodules satisfy $(d^{n+1})_L^\uparrow (d^n)_L^\uparrow = \varepsilon$, not the categorical zero map. Therefore FLR17 does not identify this support diagram with an ordinary complex in a category requiring the latter zero map. It supplies the exact fibrewise cohomology and the map recovering the original amplitude cohomology, rather than erasing supported cycles.

3 Exact matrix transport from the complete lattice base

The base in this section is the original bounded distributive lattice L , with its actual bottom 0_L and top 1_L , and the complete semiring $S_L = G_L(\mathbb{C})$. It is not a two-element replacement. The source is version 11 of *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, by The Clankers, Definition `def:lattice-split`, Theorems `thm:lattice-normal-form`, `thm:lattice-monomial-gl` and `thm:lattice-split-determinant`. Those constructions and the monomial theorem belong to that source; they are not claimed here as new results. The calculation below applies that base to the exact ST source, metric, observation and prime actions, and constructs the carriers on which their complete support transport is valid.

3.1 The amplitude and support matrix together

For an $m \times n$ matrix over S_L write (T, A) , where $T \in M_{m,n}(\mathbb{C})$ and $A \in M_{m,n}(L)$. The admissibility condition and operations are exactly

$$\begin{aligned} T_{ij} \neq 0 &\implies A_{ij} = 1_L, \\ (T, A) + (U, B) &= (T + U, A \vee B), \quad (T, A)(U, B) = (TU, A \star B), \\ (A \star B)_{ij} &= \bigvee_{\ell} (A_{i\ell} \wedge B_{\ell j}). \end{aligned} \quad (\text{CLM1})$$

The joins in the second line of CLM1 are entrywise. Each formula follows by applying the componentwise scalar operations in the cited definition to the finite matrix sum. Conversely every admissible pair specifies exactly one S_L -matrix. In particular these formulas retain complex signs in the amplitude and every lattice element in the support matrix.

Let $\beta(T)_{ij} = 1_L$ when $T_{ij} \neq 0$ and 0_L otherwise, and put $\ell(T) = (T, \beta(T))$. This is the minimal entrywise lift: a zero entry is actual $\tau_L = (0, 0_L)$. The exact failures of addition and composition are

$$\begin{aligned} \ell(T) + \ell(U) &= (T + U, \beta(T) \vee \beta(U)), & \beta(T + U) &\leq \beta(T) \vee \beta(U), \\ \ell(T)\ell(U) &= (TU, \beta(T) \star \beta(U)), & \beta(TU) &\leq \beta(T) \star \beta(U). \end{aligned} \quad (\text{CLM2})$$

For the first inequality a nonzero sum requires a nonzero summand. For the second a nonzero matrix entry requires at least one nonzero product in its defining sum. Each inequality is therefore entrywise in L . At an entry where the amplitude cancels but the right support is nonzero, the exact result is the zero-fibre point $(0, \lambda)$ with that support, not τ_L . No subtraction of lattice values is used or needed.

More generally all lifts of a fixed amplitude T form the explicit set

$$\mathfrak{F}_L(T) = \{A \in M_{m,n}(L) : T_{ij} \neq 0 \Rightarrow A_{ij} = 1_L\}. \quad (\text{CLM3})$$

CLM1 gives the composition map $\mathfrak{F}_L(T) \times \mathfrak{F}_L(U) \rightarrow \mathfrak{F}_L(TU)$, $(A, B) \mapsto A \star B$. The proof is the same nonzero-product argument. Thus cancelled matrix coefficients define actual fibres of support data, with composition, rather than invalidating the base.

3.2 The original nilpotent and projector equations

Fix the original coordinate basis and metric G_N of ST1–ST3. The following symbols are the original matrices, not orthogonally changed coordinates:

$$\begin{aligned} J^\dagger J &= I_B, & P &= J J^\dagger = L\Lambda, & P^2 &= P, & \Pi &= ee^\dagger + ff^\dagger, & \Pi^2 &= \Pi, \\ R &= \epsilon f e^\dagger, & e^\dagger f &= 0, & \Pi R &= R \Pi = R, & R^2 &= 0, & e^\dagger &= e^* G_N. \end{aligned} \quad (\text{CLM4})$$

The lattice L and the original minimum section L in $P = L\Lambda$ are distinguished by their arguments; no source section is altered. For clarity the support matrices below always carry a subscript or appear as the second member of a matrix pair.

Choose any exact lifts $\hat{R} = (R, A_R)$, $\hat{\Pi} = (\Pi, A_\Pi)$ and $\hat{P} = (P, A_P)$. CLM1 gives

$$\begin{aligned} \hat{R}^j &= (0, A_R^{*j}) \quad (j \geq 2), & \hat{\Pi}^j &= (\Pi, A_\Pi^{*j}) \quad (j \geq 1), & \hat{P}^j &= (P, A_P^{*j}) \quad (j \geq 1), \\ \hat{\Pi} \hat{R} &= (R, A_\Pi \star A_R), & \hat{R} \hat{\Pi} &= (R, A_R \star A_\Pi). \end{aligned} \quad (\text{CLM5})$$

These identities prove the exact zero-amplitude support left by the nilpotent equation and the exact support iterations left by projector equations. They do not assert nilpotence or idempotence with actual τ_L at every vanished amplitude entry.

For the minimal lift of the original rank-one R this calculation is explicit in its actual coordinates. Put $b_j = (e^* G_N)_j$ and let F_i be 1_L or 0_L according as $f_i \neq 0$ or $f_i = 0$; define B_j from b_j in the same way. Then

$$\begin{aligned} \beta(R)_{ij} &= F_i \wedge B_j, & \omega &= \bigvee_\ell (B_\ell \wedge F_\ell), & \beta(R)_{ij}^{*2} &= F_i \wedge \omega \wedge B_j, \\ \ell(R)^2 &= \begin{cases} (0, 0), & \omega = 0_L, \\ (0, \beta(R)), & \omega = 1_L. \end{cases} \end{aligned} \quad (\text{CLM6})$$

The first equality follows from $R_{ij} = \epsilon f_i b_j$ and $\epsilon > 0$. Finite distributivity proves the second line's support product. The amplitude of every entry of R^2 is $\epsilon^2 f_i (\sum_\ell b_\ell f_\ell) b_j = 0$, with the original complex terms preserved by $e^\dagger f = 0$. If $\omega = 1_L$, at least one summand $b_\ell f_\ell$ is nonzero; because their sum is zero, there are at least two nonzero summands. Thus the surviving support records a real cancellation in the original complex amplitudes. In this case $\beta(R)^{*j} = \beta(R)$ for every $j \geq 1$; when $\omega = 0_L$, its powers vanish for $j \geq 2$. No intermediate lattice level in a nonminimal lift is excluded by CLM6; those are retained in CLM5.

For the actual section, observation and adjoint the corresponding minimal products are

$$\begin{aligned} \ell(J^\dagger) \ell(J) &= (I_B, \beta(J^\dagger) \star \beta(J)), & \ell(J) \ell(J^\dagger) &= (P, \beta(J) \star \beta(J^\dagger)), \\ \ell(L) \ell(\Lambda) &= (P, \beta(L) \star \beta(\Lambda)), & \ell(ee^\dagger) + \ell(ff^\dagger) &= (\Pi, \beta(ee^\dagger) \vee \beta(ff^\dagger)). \end{aligned} \quad (\text{CLM7})$$

The two lifts of P on the first two lines need not have identical support. They have the identical original amplitude and lie in the same fibre $\mathfrak{F}_L(P)$. The adjoint support is computed from $J^\dagger = J^* G_N$; it is not replaced by the transposed support of J . These equations retain exactly the original metric coefficients responsible for cancellation.

3.3 Finite path support and the carrier that recovers an exponential group

Let (T, A) be any square admissible pair of size $n \geq 1$, over any bounded distributive L . Define its finite path closure

$$A^* = I_L \vee A \vee A^{*2} \vee \dots \vee A^{*(n-1)}. \quad (\text{CLM8})$$

Here I_L has top on its diagonal and bottom elsewhere. A matrix entry A_{ij} labels the arrow from input index j to output index i . An entry $(A^{*m})_{ij}$ is the join of the meets of edge labels along all length- m directed walks from column index j to row index i . This is proved by induction on the walk length m using finite distributivity. If a walk repeats a vertex, deleting the intervening closed part leaves the endpoints unchanged and removes some factors from a meet; the remaining meet is greater than or equal to the original one. A walk with distinct endpoints reduces to a simple path of length at most $n - 1$. A closed walk is bounded by the diagonal entry of I_L . Consequently

$$A^{*j} \leq A^* \quad (j \geq 0), \quad A^* \star A^* = A^*, \quad I_L \leq A^*. \quad (\text{CLM9})$$

For the product equality, concatenate the two finite paths in every term of $A^* \star A^*$ and use the preceding deletion argument to obtain the upper bound. The diagonal unit term gives the reverse bound. Thus no completeness assumption on L or infinite lattice join is required.

For every $z \in \mathbb{C}$, the pair

$$\mathcal{E}_A(z) = (e^{zT}, A^*) \quad (\text{CLM10})$$

is an admissible S_L -matrix. Indeed a nonzero amplitude entry of the entire exponential has a nonzero summand from some power T^j , or from I when $j = 0$. A nonzero power entry has a path with nonzero amplitude product; every edge on that path has support 1_L in A . Deleting cycles retains top support, so A^* has top in that entry. This proves admissibility. The support in CLM10 is the finite join recording every possible path, including paths whose amplitude sums vanish; it is not the minimal support of the already-summed matrix e^{zT} .

This path support is also the exact eventual support of the actual finite exponential sums. For $z \neq 0$, every coefficient $z^j/j!$ is nonzero and has supported scalar lift $(z^j/j!, 1_L)$. Therefore

$$\sum_{j=0}^D (z^j/j!, 1_L)(T, A)^j = \left(\sum_{j=0}^D \frac{z^j T^j}{j!}, \bigvee_{j=0}^D A^{*j} \right).$$

By CLM9 the support is exactly A^* for every $D \geq n - 1$, while the amplitude converges to the original entire exponential. Thus the limit requires only ordinary finite-dimensional complex convergence and eventually constant lattice coordinates, not a new infinite-sum operation on L . At $z = 0$, using minimal coefficient lifts would give (I, I_L) instead. CLM10 keeps (I, A^*) at that parameter as the exact identity of the recovered carrier in CLM11, with its retained zero-amplitude paths made explicit.

Define $P_A = (I_n, A^*)$ and the free S_L -semimodule $\mathcal{V}_L = S_L^n$. Its exact image and exponential law are

$$\begin{aligned} P_A^2 &= P_A, & \mathcal{V}_A &= \text{im } P_A = \{(v, \lambda) \in S_L^n : A^* \lambda = \lambda\}, \\ \mathcal{E}_A(z) \mathcal{E}_A(w) &= \mathcal{E}_A(z + w), & \mathcal{E}_A(0) &= P_A, & \mathcal{E}_A(z) P_A &= P_A \mathcal{E}_A(z) = \mathcal{E}_A(z). \end{aligned} \quad (\text{CLM11})$$

Vectors on the first line use amplitude $v \in \mathbb{C}^n$ and support $\lambda \in L^n$, with $v_i \neq 0 \Rightarrow \lambda_i = 1_L$; $(A^* \lambda)_i = \bigvee_j (A_{ij}^* \wedge \lambda_j)$. The idempotence in CLM9 proves the first line, since the map P_A does not change v . CLM1 and $A^* \star A^* = A^*$ prove the second line. It follows that $\mathcal{E}_A$ is an exact additive-group action on the concrete retract $\mathcal{V}_A$, with inverse $\mathcal{E}_A(-z)$ there. On the full free module its product at $z, -z$ is P_A , not generally the identity. This computes the entire inverse-cancellation defect and gives the carrier on which the inverse exists.

For real $a \geq 0$ one can also use the identity-supported value at $a = 0$ and the path-supported value (e^{iaT}, A^*) for $a > 0$ to obtain a semigroup on the full free module. Its prime powers agree

exactly because $m \log p > 0$ for $m \geq 1$. Extending that particular full-module semigroup to negative arguments fails at the displayed product $\mathcal{E}_A(ia)\mathcal{E}_A(-ia) = P_A$. The retract in CLM11 supplies the full group with all masks retained, instead of dropping this defect.

The support fibre of the retract is exactly

$$\ker(p : \mathcal{V}_A \rightarrow \mathbb{C}^n) = \{(0, \lambda) : A^* \lambda = \lambda\}. \quad (\text{CLM12})$$

It is closed under coordinatewise joins and scalar meets: matrix multiplication by A^* preserves each by distributivity. The amplitude map is surjective, since the all-top support vector is fixed by A^* . Thus this is an explicit retained lattice of zero-amplitude objects, not an omitted kernel or a new complex vector space.

3.4 The original companion graph forces the synchronized carrier

Return to the original source

$$\begin{aligned} E &= \mathbb{C}[y]/Q_k, & q &= (k+1)^2, \quad k \geq 17, \quad k \equiv 1 \pmod{4}, \\ Q_k(y) &= \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], & \delta, \gamma &> 0, & M[f] &= [yf]. \end{aligned} \quad (\text{CLM13})$$

In the original monomial coordinates $1, y, \dots, y^{q-1}$, write $Q_k(y) = y^q + \sum_{j=0}^{q-1} c_j y^j$. The matrix of M has $M_{j+1,j} = 1$ for $0 \leq j < q-1$ and $M_{j,q-1} = -c_j$. None of the roots of Q_k is zero: k is odd, so each of $2a-k$ and $2b-k$ is a nonzero odd integer. Hence $c_0 = Q_k(0) \neq 0$. The nonzero amplitude edges therefore contain the entire directed cycle

$$0 \longrightarrow 1 \longrightarrow \dots \longrightarrow q-1 \longrightarrow 0, \quad \beta(M)^* = \mathbf{1}_{q \times q}. \quad (\text{CLM14})$$

Every pair of vertices is joined along this cycle in at most $q-1$ steps, so its path support is top; this proves the equality. Every other admissible support matrix for the same original M lies above $\beta(M)$ and below the all-top matrix, and thus has the same path closure. The conclusion is independent of the unspecified supports assigned to zero entries, while preserving their intermediate masks before closure.

For the original prime action, CLM11 is consequently the carrier

$$E_L^\uparrow = \{(v, 1_L) : v \in E\} \cup \{(0, \lambda) : \lambda \in L\}. \quad (\text{CLM15})$$

Precisely, its embedding in S_L^q gives every coordinate the same support λ . To prove the identification, multiplication of a support vector by the all-top matrix gives the constant vector with value $\bigvee_j \lambda_j$. A fixed vector therefore has equal coordinates; if their common value is below top, all amplitudes must be zero. Conversely every such synchronized vector is fixed. Thus the synchronized carrier is forced by the original companion prime-action graph; it is not a replacement of L by a smaller lattice.

On this carrier the original centred and uncentred prime actions are

$$\begin{aligned} \mathcal{F}_p(v, \lambda) &= (e^{i(\log p)M} v, \lambda), \\ \tilde{\mathcal{F}}_p(v, \lambda) &= (p^{k/2} e^{i(\log p)M} v, \lambda), \\ \mathcal{F}_p^m(v, \lambda) &= (e^{im(\log p)M} v, \lambda), \quad m \in \mathbb{Z}. \end{aligned} \quad (\text{CLM16})$$

CLM11 proves the group laws. Multiplication by the supported nonzero scalar $(p^{k/2}, 1_L)$ gives the uncentred formula. The original coordinate $s = k/2 + iy$ and every prime phase are retained. For the original cardinal vector at $y_{ab} = (2b-k)\gamma - i(2a-k)\delta$, the centred eigenvalue is exactly $p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}$, and the uncentred one has the additional factor $p^{k/2}$. This follows by applying the entire exponential to $M\ell_{ab} = y_{ab}\ell_{ab}$.

The lattice fibre has a separate exact statement:

$$\mathcal{F}_p(0, \lambda) = (0, \lambda) = (a, 1_L)(0, \lambda) \quad (a \in \mathbb{C}^\times). \quad (\text{CLM17})$$

Thus a semimodule eigenvector definition that admits non-absent vectors of zero amplitude assigns these points to every supported nonzero scalar eigenvalue. They cannot be counted as new numerical prime eigenvalues. Requiring nonzero amplitude recovers exactly the original complex eigenvalue relation, while CLM17 retains the entire extra fixed lattice fibre as objects.

3.5 Recovering the exact maps and their original metric

For any original complex linear map $T : V \rightarrow W$ between positive dimensional source or receiver spaces define

$$T^\dagger : V_L^\dagger \longrightarrow W_L^\dagger, \quad T^\dagger(v, \lambda) = (Tv, \lambda). \quad (\text{CLM18})$$

These maps preserve the S_L -semimodule operations. In detail, addition is $(v, \lambda) + (w, \mu) = (v+w, \lambda \vee \mu)$ and scalar multiplication is $(r, \nu)(v, \lambda) = (rv, \nu \wedge \lambda)$. The linear identities $T(v+w) = Tv + Tw$ and $T(rv) = rTv$ prove both assertions, including every zero-amplitude case. Formula CLM18 is represented on the synchronized carriers by the matrix pair $(T, \mathbf{1})$, even when some entries of T vanish.

Direct evaluation proves

$$(TU)^\dagger = T^\dagger U^\dagger, \quad I^\dagger = I_{V_L^\dagger}, \quad (T+U)^\dagger = T^\dagger + U^\dagger. \quad (\text{CLM19})$$

An amplitude isomorphism therefore has the exact inverse $(T^{-1})^\dagger$ on this carrier. This does not assert an inverse for its matrix on the full free S_L -module, which would contradict the monomial theorem quoted from version 11. The image of the amplitude zero map is specifically

$$Z_V = 0^\dagger : (v, \lambda) \longmapsto (0, \lambda). \quad (\text{CLM20})$$

It is not the actual semimodule zero morphism $(v, \lambda) \mapsto (0, 0_L)$. In particular the recovered nilpotent identity is $(R^\dagger)^2 = Z_E$. It retains every zero layer; it does not falsely assert square-zero with actual τ_L as every output. The preimage of actual τ_L under $T^\dagger$ consists only of actual τ_L , while its layerwise kernel is the exact subset

$$\{(v, 1_L) : Tv = 0\} \cup \{(0, \lambda) : \lambda \in L\} = (\ker T)_L^\dagger.$$

This follows by imposing the output amplitude equation $Tv = 0$ without deleting the output support. Thus the distinction between categorical zero and the support-preserving zero map is explicit.

The original metric has an exact lift, not a changed amplitude metric:

$$h_{G_N}^\dagger((v, \lambda), (w, \mu)) = (v^* G_N w, \lambda \wedge \mu) \in S_L. \quad (\text{CLM21})$$

If $v^* G_N w \neq 0$, both vectors have nonzero amplitude and hence both supports are top; so the pair is admissible. Conjugation on S_L is $(r, \nu) \mapsto (\bar{r}, \nu)$. Hermitian symmetry follows from that of G_N , and sesquilinearity follows from distributivity in L and sesquilinearity of the amplitude form. For example the support of $h^\dagger(x_1 + x_2, y)$ is $(\lambda_1 \vee \lambda_2) \wedge \mu = (\lambda_1 \wedge \mu) \vee (\lambda_2 \wedge \mu)$; scalar multiplication gives the analogous meet formula. Its norm at $(0, \lambda)$ is $(0, \lambda)$, which records the zero layer. Its amplitude is exactly the original squared norm; no positive complex norm for a new zero-amplitude vector has been invented.

For the original metric adjoint $T^\dagger$ between the two original metric spaces, the support meets on both sides are identical and the amplitudes obey the adjoint identity. Consequently

$$\begin{aligned} h_W^\dagger(T^\dagger x, y) &= h_V^\dagger(x, (T^\dagger)^\dagger y), \\ (J^\dagger)^\dagger J^\dagger &= I_B^\dagger, \quad J^\dagger (J^\dagger)^\dagger = P^\dagger = L^\dagger \Lambda^\dagger, \quad (P^\dagger)^2 = P^\dagger, \\ (\Pi^\dagger)^2 &= \Pi^\dagger, \quad \Pi^\dagger R^\dagger = R^\dagger \Pi^\dagger = R^\dagger, \quad (R^\dagger)^2 = Z_E. \end{aligned} \quad (\text{CLM22})$$

All equalities follow from CLM4 and CLM19, with CLM20 keeping the correct supported zero. The first line and $J^\dagger J = I$ show that $J^\dagger$ preserves the entire lifted form, including the lattice norms. This salvages the original sections, projectors, boundary map and metric without asserting that the two-dimensional additive support algebra in ST4 was a representation of the complete lattice base.

3.6 The prime compression identity on the recovered carriers

Let $U(z) = J^\dagger e^{zM} J$ and retain the original kernel $K = \ker \Lambda$, projection $Q = I - P$, and all their amplitudes and phases. The exact amplitude composition defect is $U(z+w) = U(z)U(w) + J^\dagger e^{zM} Q e^{wM} J$. Applying CLM19 gives

$$U(z+w)^\dagger = U(z)^\dagger U(w)^\dagger + (J^\dagger e^{zM} Q e^{wM} J)^\dagger. \quad (\text{CLM23})$$

To prove the amplitude formula, insert $I = P + Q$ between the two exponentials and use $P = J J^\dagger$. The lift preserves that finite sum exactly, including cancellations into Z_B . Thus the entire original-kernel memory from TDP6–TDP10 persists on the recovered full-lattice carrier. The support saturation has not silently made the observed action multiplicative.

For comparison, if one keeps minimal entrywise lifts on the full free module, the two sides of the amplitude identity have support matrices that are computed by CLM1–CLM3 and need not agree. Their exact relation is equality after amplitude projection and equality after restriction to the synchronized carrier using the saturated lifts. This is a proved correspondence between two concrete retained objects, not an identification of their distinct free support masks.

Finally, the cited lattice determinant theorem evaluates every actual square matrix pair as

$$\overset{\text{sp}}{\underset{L}{\det}}(T, A) = \left(\det T, \bigvee_{\sigma \in S_n} \bigwedge_i A_{i, \sigma(i)} \right). \quad (\text{CLM24})$$

The original signed determinant amplitude is unchanged, while the support is the exact matching value. In particular a saturated prime matrix has top determinant support even when a related signed amplitude determinant vanishes. This does not replace invertibility on the full free module by a determinant test: the monomial theorem still controls that question. The exact recovered inverse on the synchronized or path-saturated carrier is already constructed in CLM11 and CLM19. No new Frobenius trace formula or RH conclusion is asserted from its existence.

The full free module's mixed masks also have an exact retained correspondence with the original synchronized carrier. For the projection P_1 in CLM15 its fibre is

$$P_1^{-1}(v, \lambda) = \{(v, \mu) : \mu \in L^q, \bigvee_i \mu_i = \lambda, \quad v_i \neq 0 \Rightarrow \mu_i = 1_L\}. \quad (\text{CLM25})$$

This is obtained by evaluating the projection, so it gives both directions of the correspondence. If $v \neq 0$, necessarily $\lambda = 1_L$, and the nonzero amplitude coordinates already force the join to be top. The remaining coordinates are independently arbitrary: the fibre is canonically $L^{Z(v)}$, where $Z(v) = \{i : v_i = 0\}$, with the labelled nonzero coordinates fixed at top. If $v = 0$, the fibre is exactly $\{\mu \in [0_L, \lambda]^q : \bigvee_i \mu_i = \lambda\}$. Thus the mixed masks outside the synchronized carrier are retained as explicit fibres of its retraction, with all lattice levels and coordinate labels. Recovering a group action on the retract neither identifies those different masks nor asserts that they do not exist.

For the finite powerset lattice $L = \mathcal{P}(I)$, write $r = |I|$. These exact fibres have the following cardinalities:

$$\begin{aligned} |P_1^{-1}(v, 1_L)| &= 2^{r|Z(v)|} & (v \neq 0), \\ |P_1^{-1}(0, A)| &= (2^q - 1)^{|A|} & (A \subseteq I). \end{aligned} \quad (\text{CLM26})$$

The first equality follows from the exact fibre $L^{Z(v)}$ in CLM25: each labelled zero coordinate independently chooses any of the 2^r subsets of I . For the second, write the coordinate masks as $\mu_1, \dots, \mu_q \subseteq A$. Their union is A precisely when, for every atom $a \in A$, the index set $S_a = \{i : a \in \mu_i\} \subseteq \{1, \dots, q\}$ is nonempty. The map $(\mu_i)_i \mapsto (S_a)_{a \in A}$ is a bijection onto the product, over the labelled atoms of A , of the $2^q - 1$ nonempty coordinate subsets. Its inverse is $\mu_i = \{a \in A : i \in S_a\}$. This proves the second equality without identifying different atoms or different coordinate masks. When $A = \emptyset$, the empty product has value one and its sole preimage is the all-absent mask. Thus synchronization has an explicit residual-space correspondence with every full preimage counted.

4 Exact support retractions, cohomology and prime correspondences

The coefficient object in this section is the original $G_L(\mathbb{C})$ of The Clankers, *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, version 11, Definition `def:lattice-split`; the source is retained as `sources/FULL_LATTICE_FOUNDATION_V11.tex`. The original source, metric, boundary maps and prime phases are those of ST1–3, FLR1–17 and CLM1–26. The foundational split-support construction is an antecedent. Here the exact free-to-synchronized matrix squares, a category with supported zero morphisms, its kernel and cohomology calculations, and the full fibre correspondence of the original prime maps are constructed and proved.

4.1 Definitions retained throughout

Let L be a nontrivial bounded distributive lattice and $S = G_L(\mathbb{C})$. Write its elements as (c, α) , with $c \neq 0 \Rightarrow \alpha = 1_L$, and retain

$$(c, \alpha) + (d, \beta) = (c + d, \alpha \vee \beta), \quad (c, \alpha)(d, \beta) = (cd, \alpha \wedge \beta). \quad (\text{CR1})$$

The actual scalar zero is $\tau = (0, 0_L)$; the supported zero is $e = (0, 1_L)$. These are distinct. For a complex vector space V , retain FLR1 exactly:

$$V_L^\uparrow = \{(v, 1_L) : v \in V\} \cup \{(0_V, \lambda) : \lambda \in L\}. \quad (\text{CR2})$$

Its operations are FLR2. A free vector in S^n instead has amplitude $v \in \mathbb{C}^n$ and support vector $\lambda \in L^n$, subject to $v_i \neq 0 \Rightarrow \lambda_i = 1_L$. Thus a free vector is written (v, λ) ; a synchronized vector is written (v, λ) , with one common label.

For a rectangular $m \times n$ free matrix write $\hat{T} = (T, A)$ as in CLM1. Its action is

$$\hat{T}(v, \lambda) = \left(Tv, \left(\bigvee_j A_{ij} \wedge \lambda_j \right)_{i=1}^m \right). \quad (\text{CR3})$$

All dimensions used for free matrices below are positive. The synchronized construction itself also includes $V = 0$, in which case $0_L^\uparrow = L$.

4.2 The companion path closure really is all-top

CLM8–9 are valid for every finite matrix over the stated bounded distributive lattice. Induction on path length gives the join of the meets of the edge labels along all directed walks. If a walk repeats a vertex, delete the closed subwalk. This removes factors from its meet, so its value can only increase in the lattice order. A path with different endpoints reduces to length at most $n - 1$; a closed walk is bounded by the corresponding diagonal entry of I_L . This proves, with no infinite join,

$$A^{*j} \leq A^* = I_L \vee A \vee \dots \vee A^{*(n-1)}, \quad A^* \star A^* = A^*. \quad (\text{CR4})$$

For the exact original polynomial and monomial basis of CLM13, put

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b - k)\gamma + i(2a - k)\delta] = y^q + \sum_{j=0}^{q-1} c_j y^j, \quad q = (k+1)^2. \quad (\text{CR5})$$

Here $k \geq 17$, $k \equiv 1 \pmod{4}$, and $\gamma, \delta > 0$. Every factor has a nonzero root, because both odd integers $2a - k$ and $2b - k$ are nonzero. Thus $c_0 \neq 0$. Multiplication by y has entries $M_{j+1,j} = 1$ for $0 \leq j < q-1$, and $M_{j,q-1} = -c_j$. Its nonzero entries contain the directed cycle

$$0 \longrightarrow 1 \longrightarrow \dots \longrightarrow q-1 \longrightarrow 0. \quad (\text{CR6})$$

Every ordered pair is connected by at most $q-1$ edges of this cycle. Each edge has support 1_L in every admissible lift of M . Consequently $A^* = \mathbf{1}_{q \times q}$ for every such lift A , exactly as stated in CLM14.

Let J_n denote the $n \times n$ all-top lattice matrix, and let $P_n = (I_n, J_n)$. Then

$$P_n(v, \boldsymbol{\lambda}) = (v, (\bigvee_i \lambda_i, \dots, \bigvee_i \lambda_i)). \quad (\text{CR7})$$

Its fixed vectors have all labels equal. If one amplitude coordinate is nonzero, that label must be top. If the common label is below top, every amplitude coordinate is zero. This proves exactly

$$\text{im } P_q = E_L^\uparrow$$

under the original monomial-coordinate inclusion of FLR15. No Boolean branch has been selected. Every element of L is present as a common zero-amplitude label.

The result is specific about its construction: this is the carrier of the path-supported exponential of CLM10. It does not assert that every possible lifting procedure or every mixed free mask is already that carrier.

4.3 The exact free-to-synchronized squares

Define maps

$$\begin{aligned} \iota_n : (\mathbb{C}^n)_L^\uparrow &\longrightarrow S^n, & \iota_n(v, \lambda) &= (v, (\lambda, \dots, \lambda)), \\ r_n : S^n &\longrightarrow (\mathbb{C}^n)_L^\uparrow, & r_n(v, \boldsymbol{\lambda}) &= (v, \bigvee_j \lambda_j). \end{aligned} \quad (\text{CR8})$$

Both are well defined. For r_n , a nonzero coordinate of v forces the total join to be top. Both preserve addition. Their compatibility with scalar multiplication follows from $\alpha \wedge \bigvee_j \lambda_j = \bigvee_j (\alpha \wedge \lambda_j)$. They are therefore S -linear, with

$$r_n \iota_n = I, \quad \iota_n r_n = P_n. \quad (\text{CR9})$$

For $A \in M_{m,n}(L)$, retain its exact row joins, column joins and total join:

$$\rho_i(A) = \bigvee_j A_{ij}, \quad \kappa_j(A) = \bigvee_i A_{ij}, \quad d(A) = \bigvee_{i,j} A_{ij}. \quad (\text{CR10})$$

For the original amplitude T , define $T^\uparrow(v, \lambda) = (Tv, \lambda)$, exactly as in FLR3. Direct evaluation of CR3 gives the complete two squares:

$$\begin{aligned}
\widehat{T}\iota_n(v, \lambda) &= (Tv, (\rho_i(A) \wedge \lambda)_i), \\
\iota_m T^\uparrow(v, \lambda) &= (Tv, (\lambda)_i), \\
r_m \widehat{T}(v, \lambda) &= \left(Tv, \bigvee_j \kappa_j(A) \wedge \lambda_j \right), \\
T^\uparrow r_n(v, \lambda) &= \left(Tv, \bigvee_j \lambda_j \right).
\end{aligned} \tag{CR11}$$

Consequently the restriction square commutes on its entire domain exactly when every row join is top. Necessity follows by putting $v = 0$, $\lambda = 1_L$; sufficiency follows from the first two formulas. The retraction square commutes on its entire domain exactly when every column join is top. For necessity take the zero amplitude and a support vector with top in just column j and bottom elsewhere. Sufficiency follows from the last two formulas. This proves both directions of each assertion.

For example, take $L = \{0, 1\}$, $T = \text{diag}(1, 0)$, and its minimal support $A = \text{diag}(1, 0)$. The synchronized zero-amplitude vector with label 1 is sent to the nonsynchronized mask $(1, 0)$. Thus an arbitrary minimal free lift does not restrict to the lift FLR3. This counterexample concerns that particular lift; it does not obstruct the saturated lift.

Indeed the saturated matrix $\widehat{T}_{\text{sat}} = (T, \mathbf{1}_{m \times n})$ satisfies both squares. Its exact matrix expression is

$$\widehat{T}_{\text{sat}} = \iota_m T^\uparrow r_n. \tag{CR12}$$

No amplitude, sign, coordinate or constant has changed in CR12. The support operation is explicitly specified.

Sandwiching an arbitrary lift gives the stronger exact formula

$$P_m \widehat{T} P_n = (T, d(A) \mathbf{1}_{m \times n}), \tag{CR13}$$

where $d(A) \mathbf{1}$ means that every entry is the same lattice element $d(A)$, not a numerical multiple. This follows by finite distributivity in $J_m \star A \star J_n$. If $T \neq 0$, admissibility forces some entry of A to be top, hence $d(A) = 1_L$, and CR13 equals the saturated lift. If $T = 0$, the total support $d(A)$ remains necessary data. In particular, sandwiching the minimal zero matrix gives the actual zero matrix, while the lift of the original zero linear map is supported zero. This explicitly locates the exceptional zero-map issue.

Saturation itself is the matrix operation

$$\widehat{T} \longmapsto \widehat{T} + (0, \mathbf{1}_{m \times n}). \tag{CR14}$$

Finite sums and composable products of these saturated matrices agree with the saturated matrix of the corresponding amplitude sum or product; the intermediate dimension is positive, so the support product of the two all-top rectangular matrices is all-top. Their identity on the synchronized image is P_n , as proved in CR9. These statements are exact maps, not an identification of the distinct support matrices A .

4.4 A category whose zero object is the retained support lattice

Regard L itself as an S -semimodule: its addition is join and

$$(c, \alpha) \cdot \lambda = \alpha \wedge \lambda. \tag{CR15}$$

The scalar laws follow immediately from CR1 and lattice distributivity. Define Ret_L as follows. An object is an S -semimodule X with specified S -linear maps

$$L \xrightarrow{s_X} X \xrightarrow{\sigma_X} L, \quad \sigma_X s_X = I_L. \quad (\text{CR16})$$

A morphism $f : X \rightarrow Y$ is S -linear and satisfies

$$f s_X = s_Y, \quad \sigma_Y f = \sigma_X. \quad (\text{CR17})$$

Composition and identities preserve these equations, so this defines a category without any additional existence assumption. The object (L, I_L, I_L) is both initial and terminal: a morphism from it to X must be s_X , and a morphism from X to it must be σ_X . Thus its zero morphism from X to Y is exactly

$$0_{X,Y}^{\text{Ret}} = s_Y \sigma_X. \quad (\text{CR18})$$

The two composition laws for these zero morphisms follow from CR17. This is a supported zero, not the constant map to actual τ .

The synchronized object CR2 has

$$s_V(\lambda) = (0_V, \lambda), \quad \sigma_V(v, \lambda) = \lambda. \quad (\text{CR19})$$

These maps are S -linear by FLR2 and satisfy CR16. The lift of every original complex linear map is a morphism in Ret_L . Moreover every morphism $f : V_L^\uparrow \rightarrow W_L^\uparrow$ in this category is the lift of exactly one original linear map. To prove this, CR17 forces $f(0, \lambda) = (0, \lambda)$ and forces every output to have its input label. Write

$$f(v, 1_L) = (T(v), 1_L).$$

Additivity of f gives $T(v + w) = T(v) + T(w)$; scalar multiplication by $(c, 1_L)$, together with $T(0) = 0$, gives $T(cv) = cT(v)$ for every $c \in \mathbb{C}$. Thus T is complex linear and $f = T^\uparrow$. Its uniqueness follows by reading the amplitude at label top. This proves full faithfulness, not only faithfulness, of the synchronized recovery in the specified category.

In particular FLR5 and CLM20 are the actual zero morphisms of Ret_L . FLR4 and CLM19 preserve the zero object and zero morphisms in this category. No claim about preservation of the different zero object in the ambient category of all S -semimodules is needed or made.

The free module S^n also has a specified object of Ret_L :

$$s_n(\lambda) = (0, (\lambda)_i), \quad \sigma_n(v, \lambda) = \bigvee_i \lambda_i. \quad (\text{CR20})$$

CR8 gives a retraction and inclusion in this category. CR11 proves that a free matrix (T, A) is a morphism between these free objects exactly when all its row joins and all its column joins are top. Every path-closure support A^* has this property because its diagonal is top. Saturated lifts have it as well. Thus the path-supported exponential and the saturated original maps live in one explicitly constructed category, alongside the free objects and their retained mixed masks.

The zero endomorphism of the free object CR20 is

$$0_{S^n, S^n}^{\text{Ret}} = (0, J_n) = s_n \sigma_n. \quad (\text{CR21})$$

It is different both from the actual zero matrix and, on nonsynchronized masks, from the diagonal matrix eI_n . On the synchronized image it is exactly FLR5.

4.5 Exact kernels, cokernels and cohomology

Let $T : V \rightarrow W$ be an original linear map. The inclusion

$$(\ker T)_L^\uparrow \hookrightarrow V_L^\uparrow \quad (\text{CR22})$$

is its categorical kernel in Ret_L , not merely an inverse image selected after forgetting the category. Indeed the equation $T^\uparrow f = 0_{X,W}^{\text{Ret}}$ for a morphism $f : X \rightarrow V_L^\uparrow$ says exactly that every amplitude of $f(x)$ lies in $\ker T$. Its support remains $\sigma_X(x)$. It therefore factors uniquely through CR22; the factor is S -linear and preserves both maps CR19 because CR22 is an inclusion with inherited operations. This verifies the full kernel universal property for every object X of Ret_L .

The original quotient induces

$$q^\uparrow : W_L^\uparrow \longrightarrow (W/\text{im } T)_L^\uparrow. \quad (\text{CR23})$$

It is a categorical cokernel in the same category. Its composite with $T^\uparrow$ is CR18. For the universal property let $f : W_L^\uparrow \rightarrow X$ satisfy $fT^\uparrow = 0^{\text{Ret}}$. At label top and $v \in V$,

$$f(w + Tv, 1_L) = f(w, 1_L) + f(Tv, 1_L) = f(w, 1_L) + s_X(1_L) = f(w, 1_L). \quad (\text{CR24})$$

The last equality follows by writing $s_X(1_L) = f(0_W, 1_L)$ and using additivity of f . Therefore the definition $\bar{f}([w], 1_L) = f(w, 1_L)$, $\bar{f}(0, \lambda) = s_X(\lambda)$ is independent of representatives and consistent at $([0], 1_L)$. It factors f . Since $q^\uparrow$ is surjective and S -linear, the factor is S -linear; its two support identities follow from those of f . Surjectivity also proves uniqueness. This proves CR23 completely.

It follows from CR22–23 that the categorical image of $T^\uparrow$, computed as the kernel of its cokernel, is exactly $(\text{im } T)_L^\uparrow$. Thus no part of the amplitude image or zero lattice is omitted.

For any original complex (V^n, d^n) , its lifted differentials satisfy

$$(d^{n+1})^\uparrow (d^n)^\uparrow = 0_{V^n, V^{n+2}}^{\text{Ret}}. \quad (\text{CR25})$$

They consequently form an actual complex in the pointed category just defined. The image of the preceding differential lies in the kernel of the following one by the original equation $d^n d^{n-1} = 0$, and CR22–24 compute its quotient exactly:

$$H_{\text{Ret}_L}^n(V_L^{\bullet\uparrow}) = \left(\ker d^n / \text{im } d^{n-1} \right)_L^\uparrow. \quad (\text{CR26})$$

The quotient here means the categorical cokernel of the already constructed inclusion of image into kernel. Thus CR26 proves the same object as FLR17 by exact universal properties. It strengthens FLR17's fibrewise calculation; it does not falsely turn d^2 into the constant actual-zero map in the different category of all semimodules.

The split sum in FLR11 is also the categorical product in Ret_L . The product of two objects in this category is the fibre product of their support maps, with section $\lambda \mapsto (s_V \lambda, s_W \lambda)$. For synchronized objects this is exactly $(V \oplus W)_L^\uparrow$, as FLR11 proves. Both projections and the paired map preserve support and section, verifying the product universal property. The ordinary vector-space inclusions into the direct sum lift and give its coproduct property: for morphisms $f : V_L^\uparrow \rightarrow X$, $g : W_L^\uparrow \rightarrow X$, the map

$$((v, w), \lambda) \longmapsto f(v, \lambda) + g(w, \lambda) \quad (\text{CR27})$$

is S -linear, preserves the section because $s_X \lambda + s_X \lambda = s_X \lambda$, and preserves support because $\lambda \vee \lambda = \lambda$. It restricts to f and g : for example $f(v, \lambda) + s_X \lambda = f(v, \lambda) + f(0, \lambda) = f(v, \lambda)$. Every element is the sum of the two inclusion images at its same label, so uniqueness follows. Hence the synchronized objects have the original biproduct and exact linear algebra in this category.

4.6 The metric really restricts from the full free module

Let G be the original positive definite Hermitian matrix on $\mathbb{C}^n$, in its original coordinates. Take any admissible support matrix B for G . Every diagonal entry of G is positive, so

$$B_{ii} = 1_L \quad \text{for every } i. \quad (\text{CR28})$$

Choose B symmetric to give a Hermitian free pairing; the minimal support $\beta(G)$ is one such choice. The exact free pairing is

$$\widehat{h}_B((v, \lambda), (w, \mu)) = \left(v^* G w, \bigvee_{i,j} \lambda_i \wedge B_{ij} \wedge \mu_j \right). \quad (\text{CR29})$$

This is the matrix pairing over S , with conjugation $(c, \alpha) \mapsto (\bar{c}, \alpha)$. Hermitian symmetry and sesquilinearity follow from matrix multiplication CR1, symmetry of B , and Hermitian symmetry of G . On the synchronized inclusion, its support is

$$\bigvee_{i,j} \lambda \wedge B_{ij} \wedge \mu = \lambda \wedge \mu, \quad (\text{CR30})$$

because CR28 forces the total join of B to be top. Thus FLR7 and CLM21 are the exact restriction of this free pairing, with the original amplitude metric unchanged.

Moreover P_n is self-adjoint for CR29. Each row and each column of B has join top by CR28, so both $\widehat{h}_B(P_n x, y)$ and $\widehat{h}_B(x, P_n y)$ have support $(\bigvee_i \lambda_i) \wedge (\bigvee_j \mu_j)$, and their amplitudes agree because the amplitude of P_n is I_n . Its idempotence was already proved in CR9. In general P_n is not an isometry of the full support-valued pairing: CR29 may contain strictly less support before projection. This retains, rather than suppresses, the free mixed-mask metric information.

For original positive definite metrics G_V, G_W , let $T^\dagger$ be the actual metric adjoint. Their saturated lifts are an adjoint pair even on the full free modules with pairings CR29. Both sides of

$$\widehat{h}_{B_W}(\widehat{T}_{\text{sat}} x, y) = \widehat{h}_{B_V}(x, \widehat{T}^\dagger_{\text{sat}} y) \quad (\text{CR31})$$

have the original adjoint amplitude and support $(\bigvee_i \lambda_i) \wedge (\bigvee_j \mu_j)$, again because every row and column join of the metric supports is top. Restricting CR31 through CR8 proves FLR8 and CLM22 as exact inherited pairings. No replacement of the weighted adjoint by a transposed support matrix has occurred.

In particular the exact identities involving the original section, J , P_B , Π and R follow from the amplitude identities and CR12–14. In the recovered category, $(R^\dagger)^2$ is precisely the zero morphism CR18. On the full free module its saturated counterpart squares to CR21, as the original amplitude $R^2 = 0$ requires.

4.7 Why the original prime group cannot act invertibly on every free mask

The difference between the retract and the free object can be proved for the actual original prime matrices, rather than left as a possible issue.

Here a monomial matrix has exactly one nonzero entry in every row and every column. An S -linear automorphism of S^n has a monomial amplitude matrix. Write an inverse pair as $(T, A)(U, B) = (I, I_L)$ and $(U, B)(T, A) = (I, I_L)$. The amplitude $U = T^{-1}$ is invertible. If $T_{ij} \neq 0$, then $A_{ij} = 1_L$. For every $\ell \neq i$, the off-diagonal support equation in $A \star B = I_L$ forces $B_{j\ell} = 0_L$, hence $U_{j\ell} = 0$. If a row i of T had two nonzero entries at $j \neq j'$, rows j, j' of U would both be supported on column i , contradicting invertibility of U . Thus each row of T has at most one nonzero entry. Invertibility gives exactly one per row and per column. This proves the assertion directly from CR1.

Now retain the actual companion matrix M and let

$$F_p = \exp(i(\log p)M), \quad p \text{ an ordinary prime.} \quad (\text{CR32})$$

Its amplitude is not monomial in the original monomial basis. To prove this, first observe that Q_k has at least two nonzero coefficients below its leading term. One is $c_0 \neq 0$. If all the others were zero, $Q_k = y^q + c_0$, all of whose roots have the same absolute value. But the original grid contains roots with absolute values

$$k\sqrt{\gamma^2 + \delta^2} \quad \text{and} \quad \sqrt{\gamma^2 + \delta^2}, \quad (\text{CR33})$$

using respectively $(2a - k, 2b - k) = (k, k)$ and $(1, 1)$. These are different because $k \geq 17$. This proves the coefficient claim without imposing an extra relation between γ and δ .

Every matrix commuting with M is multiplication by its value on the cyclic vector 1: on the basis element $y^j = M^j 1$, commutation gives $T(y^j) = M^j T(1)$. If such an invertible matrix is monomial, then $T(1) = cy^r$ with $c \neq 0$ and $0 \leq r < q$. If $r \geq 1$, take $j = q - r$, which is an available basis index. Its image is

$$T(y^{q-r}) = cy^q = -c \sum_{j=0}^{q-1} c_j y^j, \quad (\text{CR34})$$

which has at least two nonzero coordinates by the preceding coefficient calculation. This contradicts the monomial property. Hence $r = 0$ and such a commuting monomial matrix must be cI .

But F_p commutes with M and is not scalar. The original roots are distinct because $\gamma, \delta > 0$, and the cardinal vectors give eigenvalue absolute values

$$|\exp(i(\log p)y_{ab})| = p^{(2a-k)\delta}. \quad (\text{CR35})$$

These values vary with a , so a scalar matrix is impossible. Therefore F_p is not monomial. The preceding direct automorphism proof now shows that no S -linear automorphism of the full free module can have amplitude F_p . Multiplication by the retained nonzero scalar factor $p^{k/2}$ does not change this conclusion for the uncentred prime action.

There is an even stronger finite-lattice statement. If $2 \leq |L| < \infty$, no amplitude-preserving bijection of the entire free set can cover F_p . Since the invertible F_p is not monomial, one of its columns has at least two nonzero entries. Choose the corresponding basis vector e_j . The fibre over e_j has size $|L|^{q-1}$, whereas the fibre over $F_p e_j$ has size $|L|^{q-r}$ for some $r \geq 2$, by FLR16. These cardinalities differ. A bijection covering the invertible amplitude map would give a bijection of those fibres, which is impossible. This proves the claim even without linearity.

The exact positive result is the group action on $\text{im } P_q = E_L^\uparrow$ of CLM16, with retraction fibres exactly CLM25–26. The full free path-supported operators collapse those fibres to the synchronized output. For example, already with $L = \{0, 1\}$ and two coordinates, the different zero-amplitude masks $(1, 0)$ and $(1, 1)$ have the same image under P_2 . Therefore no inverse on the entire free module can be inferred from the inverse on the retract.

4.8 The exact fibre groupoid and invertible prime correspondence

Retain every free mask as an object. Put $X = S^q$, $Y = E_L^\uparrow$, $r = r_q$, and $\iota = \iota_q$. Define the groupoid $\mathcal{G}_r$ with object set X , with one arrow $x \rightarrow x'$ exactly when $r(x) = r(x')$, and with no arrow otherwise. The composite of two composable arrows is the unique arrow with the resulting endpoints. Such an arrow exists by transitivity of equality of r -images. The unique arrow $x \rightarrow x$ is the identity and reversal is the inverse. This proves the groupoid laws.

Let $\mathcal{D}(Y)$ be the discrete groupoid on the set Y . The maps r, ι define functors

$$\mathcal{R} : \mathcal{G}_r \longrightarrow \mathcal{D}(Y), \quad \mathcal{I} : \mathcal{D}(Y) \longrightarrow \mathcal{G}_r, \quad \mathcal{R}\mathcal{I} = I_{\mathcal{D}(Y)}. \quad (\text{CR36})$$

An arrow of $\mathcal{G}_r$ is sent by $\mathcal{R}$ to the identity of its common r -image. The other functor sends each identity to the identity of its included object. Their composite $\mathcal{P} = \mathcal{IR}$ acts on objects by the exact synchronization P_q of CR7. There is a specified natural isomorphism

$$\epsilon : \mathcal{P} \Longrightarrow I_{\mathcal{G}_r}, \quad \epsilon_x = (P_q x \longrightarrow x). \quad (\text{CR37})$$

The arrow exists because $r(P_q x) = r(x)$, and is invertible by the groupoid definition. For every arrow $x \rightarrow x'$, both composites in its naturality square are the unique arrow $P_q x \rightarrow x'$. This proves naturality, including all zero-amplitude masks.

For $a \in \mathbb{Q}_{>0}^\times$, use the actual recovered prime-parameter map

$$F_a^\uparrow(v, \lambda) = (\exp(i(\log a)M)v, \lambda). \quad (\text{CR38})$$

The real logarithm on positive rationals satisfies $\log(ab) = \log a + \log b$. The two matrix exponents commute because both are multiples of the same M ; their series product gives $F_a^\uparrow F_b^\uparrow = F_{ab}^\uparrow$, with identity $F_1^\uparrow = I_Y$ and inverse $F_{a^{-1}}^\uparrow$. This retains F_p for every ordinary prime. These bijections are functors of the discrete groupoid. Define

$$\mathcal{T}_a = \mathcal{IF}_a^\uparrow \mathcal{R}. \quad \mathcal{T}_a(x) = \iota F_a^\uparrow r(x). \quad (\text{CR39})$$

On arrows, this functor sends the unique arrow between two objects with common r -image to the identity of their common image under the object map in CR39. The exact composition and inverse equations are

$$\mathcal{T}_a \mathcal{T}_b = \mathcal{T}_{ab}, \quad \mathcal{T}_1 = \mathcal{P}, \quad \mathcal{T}_a \mathcal{T}_{a^{-1}} = \mathcal{P} = \mathcal{T}_{a^{-1}} \mathcal{T}_a. \quad (\text{CR40})$$

These follow by inserting $\mathcal{RI} = I$. Equation CR37 therefore supplies the explicit natural isomorphisms from both inverse composites to the identity. Each $\mathcal{T}_a$ is an equivalence of the groupoid. The composition constraints are literal identities. The unit constraints are coherent: at a synchronized object ιy , $\epsilon_{\iota y}$ is its identity; also $\mathcal{R}(\epsilon_x)$ is an identity. Thus whiskering ϵ on either side by any $\mathcal{T}_a$ yields the identity transformation required by the unit laws.

The complete mask correspondence, without choosing a preferred target mask, is

$$\mathcal{C}_a = \{(x, x') \in X \times X : r(x') = F_a^\uparrow r(x)\}. \quad (\text{CR41})$$

If composition means first $\mathcal{C}_b$ and then $\mathcal{C}_a$, then $\mathcal{C}_a \circ \mathcal{C}_b = \mathcal{C}_{ab}$. One inclusion follows by substituting the two defining equations. For the other, if $r(z) = F_{ab}^\uparrow r(x)$, choose the middle object $y = \iota F_b^\uparrow r(x)$. It satisfies both required relations. Reversal of pairs gives $\mathcal{C}_a^{-1} = \mathcal{C}_{a^{-1}}$. The relation $\mathcal{C}_1$ is equality of the r -images, not the diagonal relation on X .

These constructions retain different masks as different objects. They declare masks with the same r -image isomorphic in this particular groupoid. The equivalence on the groupoid and the relation CR41 do not contradict the absence of a free-module automorphism or a free-set bijection in the preceding nonmonomial-prime calculation. Their composite identity is explicitly the fibre relation, with the natural isomorphism CR37 supplying the categorical identity. No preservation of an analytic trace on the fibres is inferred.

5 The full additive support algebra and the original nilpotent action

The starting object is exactly the additive monoid of the source's $G_L(\mathbb{Z})$, with the join law in Definition `def:lattice-split`. Its scalar special case and original second-order action are ST4, reproduced algebraically below. Neither the foundational scalar result nor finite-poset inversion is claimed new. The calculation here determines their full-lattice receiver, every finite-dimensional extension of the original action, and its prime trace. The prime operator, source metric, and original boundary vectors remain those of ST1–3 and TDP1–4.

5.1 An exact algebra isomorphism, including every zero symbol

Write $[n]$ for the basis symbol of $(n, 1_L)$, and $[z_\lambda]$ for the symbol of $(0, \lambda)$; $[0] = [z_1]$ is a single symbol. The complex monoid algebra $\mathcal{A}_L = \mathbb{C}[G_L(\mathbb{Z}), +]$ has multiplication

$$[n][m] = [n + m], \quad [z_\lambda][n] = [n], \quad [z_\lambda][z_\mu] = [z_{\lambda \vee \mu}], \quad 1_{\mathcal{A}_L} = [z_0]. \quad (\text{FSJ1})$$

These are the original addition laws, extended bilinearly. In particular their algebra unit is the symbol of τ , not the symbol $[1]$. This specifies the operation being linearized.

For finite nontrivial L , there is the exact unital algebra isomorphism

$$\begin{aligned} \Psi : \mathcal{A}_L &\longrightarrow \mathbb{C}[t, t^{-1}] \times \mathbb{C}^{L \setminus \{1\}}, \\ [n] &\longmapsto (t^n, (0)_{a < 1}), \\ [z_\lambda] &\longmapsto (1, (\mathbf{1}_{\{\lambda \leq a\}})_{a < 1}). \end{aligned} \quad (\text{FSJ2})$$

The second and third lines agree at $[0] = [z_1]$. They preserve FSJ1 because $\lambda \vee \mu \leq a$ holds exactly when both $\lambda \leq a$ and $\mu \leq a$ hold.

Here is a full inverse. Define the finite-poset coefficients $\mu(a, a) = 1$ and $\mu(a, b) = -\sum_{a \leq c < b} \mu(a, c)$ for $a < b$. They satisfy $\sum_{a \leq c \leq b} \mu(a, c) = \mathbf{1}_{\{a=b\}}$ by this recursion. Put

$$\xi_a = \sum_{\lambda \geq a} \mu(a, \lambda) [z_\lambda] \quad (a \in L). \quad (\text{FSJ3})$$

Evaluation in the b coordinate of FSJ2 gives exactly $\sum_{a \leq \lambda \leq b} \mu(a, \lambda) = \mathbf{1}_{\{a=b\}}$. The Laurent coordinate of ξ_a is $\sum_{a \leq \lambda \leq 1} \mu(a, \lambda) = \mathbf{1}_{\{a=1\}}$. Consequently $\xi_1 = [0]$, and for $a < 1$ the element ξ_a has Laurent coordinate zero and the single scalar coordinate one at a . The inverse sends $(f, (c_a))$ to $\sum_n f_n [n] + \sum_{a < 1} c_a \xi_a$, where $f = \sum_n f_n t^n$. To check that no element was lost, write an element uniquely as $\sum_{n \neq 0} r_n [n] + \sum_\lambda b_\lambda [z_\lambda]$. Zero Laurent coordinates force $r_n = 0$ and $\sum_\lambda b_\lambda = 0$. The proper scalar coordinates successively force every $b_a = 0$ for $a < 1$, starting at the bottom and proceeding in increasing order. The remaining sum forces $b_1 = 0$. This proves injectivity as well as the displayed inverse.

Thus $\xi_a^2 = \xi_a$, $\xi_a \xi_b = 0$ for $a \neq b$, and $\sum_a \xi_a = 1$. For the Boolean lattice $L = \mathcal{P}(I)$,

$$\mu(A, B) = (-1)^{|B \setminus A|}, \quad \xi_A = \sum_{B \supseteq A} (-1)^{|B \setminus A|} [z_B]. \quad (\text{FSJ4})$$

The formula follows because the defining interval sum is $(1 - 1)^{|B \setminus A|}$ off the diagonal. In two channels, $\xi_\emptyset = [z_\emptyset] - [z_{\{1\}}] - [z_{\{2\}}] + [z_{\{1,2\}}]$ is exactly the mixed vector FLC10, now with its algebra multiplication also computed.

The ideal generated by $([1] - [0])^2$ is mapped by FSJ2 to $((t - 1)^2) \times 0$. Hence its quotient is exactly

$$\mathcal{A}_L^{(2)} = \mathcal{A}_L / (([1] - [0])^2) \simeq \mathbb{C}[\eta] / (\eta^2) \times \mathbb{C}^{L \setminus \{1\}}, \quad t \longmapsto 1 + \eta. \quad (\text{FSJ5})$$

This holds also for negative integers, since $(1 + \eta)^{-1} = 1 - \eta$ and induction gives $(1 + \eta)^n = 1 + n\eta$ for every $n \in \mathbb{Z}$. The proper support idempotents survive the jet quotient unchanged.

The inclusion of the scalar submonoid $\{\tau\} \sqcup \mathbb{Z}$ gives the injective map

$$\iota : \mathbb{C} \times \mathbb{C}[\eta] / (\eta^2) \longrightarrow \mathcal{A}_L^{(2)}, \quad (c_0, d) \longmapsto (d, (c_0)_{a < 1}). \quad (\text{FSJ6})$$

It sends the original $[\tau]$ and $[n]$ to the identical full-lattice symbols. Thus ST4 specifies the action of this embedded algebra; it does not specify the separate actions of the ξ_a , $a < 1$.

For $L = \mathcal{P}(I)$, each coordinate branch $\beta_i(z_A) = z_{\mathbf{1}_{\{i \in A\}}}$ induces a surjection $\mathcal{A}_L^{(2)} \rightarrow \mathcal{A}_{\mathbb{B}}^{(2)}$. In FSJ5 coordinates it keeps the dual-number factor and the single scalar factor $a = I \setminus \{i\}$. This is because $\mathbf{1}_{\{i \notin A\}} = \mathbf{1}_{\{A \subseteq I \setminus \{i\}\}}$. The kernel after all individual branches have been taken is therefore

$$\ker \left(\mathcal{A}_L^{(2)} \longrightarrow \prod_{i \in I} \mathcal{A}_{\mathbb{B}}^{(2)} \right) = \bigoplus_{|I \setminus A| \geq 2} \mathbb{C} \xi_A. \quad (\text{FSJ7})$$

This proves the algebraic relation to FLC7–11, not merely a match of dimensions: a branch's two zero-symbol coordinates are total mass and its marginal, and its Laurent/jet coordinate is the common amplitude coordinate. Complementing subsets interchanges join and meet and sends these ξ_A to the primitive joint-moment channels indexed by $I \setminus A$.

5.2 The complete space of extensions in the original metric

Retain the original E , G_N , $\Pi = ee^\dagger + ff^\dagger$, $R = \epsilon f e^\dagger$, and $H = \ker \Pi$. In particular $\dim E = q$, $\dim H = q - 2$, $R^2 = 0$ and $R \neq 0$. All algebra representations $\rho_L : \mathcal{A}_L^{(2)} \rightarrow \text{End}_{\mathbb{C}}(E)$ extending ST4 are exactly

$$\begin{aligned} \rho_L(d_0 + d_1\eta, (c_a)) &= d_0\Pi + d_1R + \sum_{a<1} c_a E_a, \\ E_a^2 &= E_a, \quad E_a E_b = 0 \ (a \neq b), \quad E_a \Pi = \Pi E_a = 0, \quad \sum_{a<1} E_a = I_E - \Pi. \end{aligned} \quad (\text{FSJ8})$$

For necessity take $E_a = \rho_L(\xi_a)$ and use FSJ3 and FSJ6. The complementary idempotent equals $I - \Pi$ by extension of ST4. The dual factor must be $d_0\Pi + d_1R$. Conversely the displayed relations, $\Pi R = R\Pi = R$, and $R^2 = 0$ prove multiplicativity of FSJ8 by direct multiplication; $E_a R = R E_a = 0$ follows by inserting Π next to R . The sum of the displayed idempotents proves unitality and FSJ6 proves the required extension.

For orthogonal support projections in the unchanged metric G_N , add precisely $E_a^\dagger = E_a$ to FSJ8. Such extensions are equivalently all orthogonal decompositions

$$H = \bigoplus_{a<1} H_a, \quad E_a = \text{the original-metric orthogonal projection onto } H_a. \quad (\text{FSJ9})$$

Indeed the idempotent relations make the ranges orthogonal and give their sum H ; conversely an orthogonal decomposition has exactly those projections. Every tuple of integers $r_a \geq 0$ with $\sum r_a = q - 2$ occurs: split an orthonormal basis of the given Hermitian space H into labelled blocks of these sizes. This does not alter G_N or the original vectors e, f . More explicitly, for any basis-column matrix U_a of H_a , its projection is $E_a = U_a (U_a^\dagger U_a)^{-1} U_a^\dagger$, with the dagger and all entries in the original metric. No rescaling of source coordinates is needed.

The resulting complete parameter space is the disjoint union

$$\coprod_{\substack{r_a \geq 0 \ (a<1) \\ \sum r_a = q-2}} \text{U}(H, G_N|_H) / \coprod_{a<1} \text{U}(r_a). \quad (\text{FSJ10})$$

To verify the quotient description, an original-metric unitary sends an orthonormal basis adapted to one decomposition to a basis adapted to any other decomposition of the same dimensions. Its stabilizer is exactly the product of the unitaries on the labelled summands. Thus FSJ10 describes every orthogonal support extension, including zero summands. Without the orthogonality condition, FSJ8 gives

$$\coprod_{\substack{r_a \geq 0 \ (a<1) \\ \sum r_a = q-2}} \text{GL}(H) / \coprod_{a<1} \text{GL}(r_a). \quad (\text{FSJ10a})$$

Indeed mutually annihilating idempotents summing to the identity are exactly all labelled direct-sum decompositions of H . Sending an adapted basis to another proves transitivity of $\text{GL}(H)$, and the invertible maps preserving each summand are exactly the displayed stabilizer. This proves both parameter-space descriptions.

The dual-number factor acts faithfully, since Π and R are linearly independent: apply $d_0\Pi + d_1R$ first to f and then to e . The whole FSJ8 action is faithful exactly when every $E_a \neq 0$. Therefore faithful orthogonal extensions exist exactly when

$$|L| - 1 \leq q - 2. \quad \text{For } L = \mathcal{P}(I) : \quad 2^{|I|} \leq q - 1. \quad (\text{FSJ11})$$

This follows in both directions from the dimension allocations in FSJ9. At $k = 17$, $q = 324$, eight Boolean support coordinates fit, whereas nine do not fit faithfully in this particular jet receiver. This is an exact dimension statement about the linearized support algebra, not a bound on the number of primes, a statement about the full semimodule $E_L^\uparrow$, or a contradiction to RH.

5.3 The exact additional prime-trace coefficients

For every extension in FSJ8,

$$\rho_L([z_\lambda]) = \Pi + \sum_{\lambda \leq a < 1} E_a, \quad \text{tr } \rho_L([z_\lambda]) = 2 + \sum_{\lambda \leq a < 1} r_a. \quad (\text{FSJ12})$$

The first formula is FSJ2; the second follows from the ranks of the idempotents. In particular the supported zero $[z_1]$ has trace two and the actual-absence symbol $[z_0] = 1$ has trace q in this additive monoid algebra. These traces have been derived from the specified representation; they are not assigned from the heights of prime ideals in the semiring spectrum.

Retain every cardinal ℓ_{bc} and every prime eigenvalue in TDP2–3. Write $\text{ev}_{bc} : E \rightarrow \mathbb{C}$ for evaluation at the original root λ_{bc} , and put $\omega_{bc}(p, m) = p^{m(2b-k)\delta} \exp(im(2c-k)\gamma \log p)$ for $0 \leq b, c \leq k$. Then

$$\begin{aligned} \text{tr}(\rho_L([z_\lambda])F_p^m) &= \sum_{b,c=0}^k \omega_{bc}(p, m) \left[\text{ev}_{bc}(\Pi \ell_{bc}) + \sum_{\lambda \leq a < 1} \text{ev}_{bc}(E_a \ell_{bc}) \right], \\ \text{tr}(F_p^m) &= \sum_{b,c=0}^k \omega_{bc}(p, m). \end{aligned} \quad (\text{FSJ13})$$

The notation b, c in this display occupies respectively the original first and second quartet indices; no root or phase has changed. To prove the formulas, use the exact dual basis $\text{ev}_{bc}(\ell_{de}) = \delta_{bd}\delta_{ce}$ and $F_p^m \ell_{bc} = \omega_{bc}(p, m) \ell_{bc}$. The trace of any operator T is $\sum_{bc} \text{ev}_{bc}(T \ell_{bc})$. Substitution proves the first identity. For $\lambda = 0$, FSJ8 makes the bracket equal to $\text{ev}_{bc}(\ell_{bc}) = 1$, proving the second. For the uncentred action multiply every term by exactly $p^{mk/2}$.

Thus the full original trace is recovered after summing all support channels, and FSJ13 calculates the individual coefficients it leaves undetermined before an extension is specified. The source's scalar action fixes Π, R and $\sum E_a$; FSJ10 records the orthogonal extension space and FSJ10a records all algebra extensions. No choice is inferred from the old scalar prime-chain theorem. These coefficients are also not the freely chosen place weights of FLC15: FSJ13 is a trace of original finite operators, whereas FLC15 is a specified valuation into the local distributions of the explicit formula. Both maps and the mixed kernel FSJ7 are now explicit.

6 The full lattice in the prime and adelic calculation

Sources. The coefficient definition, spectral ordinal sum, branch selection and adelic defect are from The Clankers, *Split Support Geometry*, Version 11, retained in the original TeX, especially `def:lattice-split`, `thm:spectral-ordinal-sum`, `thm:branch-selected-spectrum` and `def:adelic-support-defect`. Alain Connes's original *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, [arXiv:math/9811068v1](https://arxiv.org/abs/math/9811068v1), Section VI, equations (1)–(16), supplies the adelic action and its formal local fixed-point terms. The complete original source archive and decompressed TeX are retained. The following maps extend the calculation to G_L without identifying a formal permutation trace with Connes's analytic cohomological trace.

Let $\mathbb{A} = \mathbb{A}_{\mathbb{Q}}$, $X = \mathbb{A}/\mathbb{Q}^\times$ and $C = \mathbb{A}^\times/\mathbb{Q}^\times$. Multiplication by a rational or an idele acts on $G_L(\mathbb{A})$ by

$$u(a, 1_L) = (ua, 1_L), \quad uz_\lambda = z_\lambda. \quad (\text{FLC1})$$

These are actions of the multiplicative groups on the underlying set; they are not asserted to preserve the multiplicative unit as semiring automorphisms. They preserve each zero-support label because $(u, 1_L)(0, \lambda) = (0, \lambda)$. They commute, so taking rational orbits gives the C -set

$$X_L = G_L(\mathbb{A})/\mathbb{Q}^\times = \{z_\lambda : \lambda \in L\} \sqcup (X \setminus \{[0]\})^\bullet. \quad (\text{FLC2})$$

Every zero label is a singleton rational orbit, and every nonzero represented orbit is the original rational orbit of its amplitude. This proves the decomposition and its equivariance. The amplitude map $a_X : X_L \rightarrow X$ has fibre L at $[0]$ and a singleton at every other orbit. For $[a] \neq [0]$, the fixed-point equation is exactly

$$[j][a]^\bullet = [a]^\bullet \iff \exists q \in \mathbb{Q}^\times \quad (qj_v - 1)a_v = 0 \quad \text{at every place } v. \quad (\text{FLC3})$$

Thus the entire lattice of zero states is fixed by C , while the nonzero fixed orbits retain this equation. At a point with just one zero coordinate v , q is uniquely determined by any nonzero coordinate; putting $t_v = qj_v$ gives transverse action $a_v \mapsto t_v a_v$ and denominator $|1 - t_v|_v$. This is the original Connes local amplitude denominator, with its quotient representative handled explicitly.

Let $\mathbb{C}[Y]$ denote finite formal linear combinations of points of a set Y . The induced linear map has exact kernel

$$\ker(\mathbb{C}[X_L] \xrightarrow{a_X} \mathbb{C}[X]) = \left\{ \sum_{\lambda \in L} c_\lambda \delta_{z_\lambda} : \text{finite support, } \sum_\lambda c_\lambda = 0 \right\}. \quad (\text{FLC4})$$

Every nonzero-amplitude orbit has a unique lift, so its coefficient must vanish in the kernel; the remaining condition is the displayed sum at $[0]$. This proves FLC4. The entire kernel is a trivial C -representation. When L is finite its dimension is $|L| - 1$, and its integrated representation has trace

$$(|L| - 1) \int_C h(u) d^\times u \quad (\text{FLC5})$$

for an integrable test function h . This is a finite-dimensional trace on exactly FLC4. For infinite L no finite trace is inferred. Nor does FLC5 identify FLC4 with any specific analytic cohomology.

6.1 All scalar branches can lose mixed linear combinations

The source proves that the family of all Boolean branches separates the elements of a finite distributive lattice. Linearization is a different operation, and its exact kernel must also be computed. For $L = \mathcal{P}(I)$, $I = \{1, \dots, n\}$, $n \geq 1$, let $V_L = \mathbb{C}[L]$ with basis δ_A and define the simultaneous branch map

$$\mathcal{B} : V_L \longrightarrow \bigoplus_{i=1}^n \mathbb{C}[\{0, 1\}], \quad \delta_A \longmapsto (\delta_{\mathbf{1}_{\{i \in A\}}})_{i=1}^n. \quad (\text{FLC6})$$

Each component is precisely the linearization of the source branch at i . For coefficients c_A , its kernel is

$$\sum_A c_A = 0, \quad \sum_{A \ni i} c_A = 0 \quad (i = 1, \dots, n). \quad (\text{FLC7})$$

These equations are independent: the functions 1 and $A \mapsto \mathbf{1}_{\{i \in A\}}$ are linearly independent by evaluating first at $\emptyset$ and then at each singleton. Therefore

$$\dim \ker \mathcal{B} = 2^n - n - 1. \quad (\text{FLC8})$$

For every $T \subseteq I$ set

$$v_T = \sum_{A \subseteq I} (-1)^{|A \cap T|} \delta_A. \quad (\text{FLC9})$$

The v_T form an orthogonal basis: for $T \neq U$, choose a coordinate in their symmetric difference and pair subsets differing in that coordinate in the inner product. Their contributions cancel. Every vector has squared norm 2^n . If $|T| \geq 2$, its total coefficient sum vanishes by toggling a coordinate of T . The sum restricted to $A \ni i$ vanishes by toggling a coordinate in $T \setminus \{i\}$. Hence these $2^n - n - 1$ vectors form a basis of FLC7. This is the complete mixed kernel left after every individual branch has been observed, with no missing higher intersections.

For two channels the kernel is the explicit line

$$\mathbb{C}h, \quad h = \delta_\emptyset - \delta_{\{1\}} - \delta_{\{2\}} + \delta_{\{1,2\}}. \quad (\text{FLC10})$$

Both branch maps send h to zero. Its four support states remain distinct before linearization. In FLC4–5 this line is fixed by the idele action, so its finite-dimensional integrated trace is exactly $\int_G h_0(u) d^\times u$ for a test function h_0 . This calculation identifies a concrete mixed channel; it supplies neither its inclusion in an analytic Weil space nor a Weil sign.

Recovery is explicit. The joint moment map

$$\mathcal{J}(c)_T = \sum_{A \supseteq T} c_A \quad (T \subseteq I), \quad c_A = \sum_{T \supseteq A} (-1)^{|T \setminus A|} \mathcal{J}(c)_T \quad (\text{FLC11})$$

is invertible. Substitution gives coefficient 1 for c_A and, for $B \supsetneq A$, coefficient $\sum_{A \subseteq T \subseteq B} (-1)^{|T \setminus A|} = 0$. Thus retaining all joint support intersections repairs exactly the loss in FLC6. The coordinates in FLC11 are products of the original branch indicators, not additional chosen primes.

6.2 The rational prime coordinates of the support defect

Use the source's standard rational place set $\Omega = \{\infty\} \sqcup \mathbb{P}$ and its counting measure with the usual absolute values $|p|_p = p^{-1}$, $|p|_\infty = p$. For a bounded real weight $c = (c_v)_{v \in \Omega}$, the exact source defect transform is

$$\mathcal{D}(c)(a) = \sum_v c_v \log |a|_v. \quad (\text{FLC12})$$

Every rational $a \neq 0$ has only finitely many nonzero finite-place terms. Writing $a = \pm \prod_p p^{\nu_p(a)}$ gives

$$\boxed{\mathcal{D}(c)(a) = \sum_p \nu_p(a) (c_\infty - c_p) \log p.} \quad (\text{FLC13})$$

Indeed the infinite-place term is $c_\infty \sum_p \nu_p(a) \log p$ and the finite term at p is $-c_p \nu_p(a) \log p$. Thus $\ker \mathcal{D} = \mathbb{R}\mathbf{1}$: evaluation at $a = p$ forces $c_p = c_\infty$ for every prime, and these equalities conversely make every term zero. On support indicators $c_v = \mathbf{1}_A(v)$ this recovers the source's full/empty balanced support theorem, now with each prime coordinate of its defect displayed. In particular a selected finite-prime support generally changes the product-formula defect; it is not made invisible by preserving the residue field at that prime.

6.3 Exact map from place support to the prime distribution

Use Connes's original-author TeX *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022](#), section `sectweilpos`, equations `wp` and `arch`, for the following explicit-formula convention. For $f \in C_c^\infty(\mathbb{R}_+^\times)$ put

$$\begin{aligned} W_p(f) &= (\log p) \sum_{m \geq 1} p^{-m/2} (f(p^m) + f(p^{-m})), \\ W_{\text{arch}}(f) &= (\log(4\pi) + \gamma) f(1) \\ &\quad + \int_1^\infty [f(x) + f(x^{-1}) - 2x^{-1/2} f(1)] \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}. \end{aligned} \quad (\text{FLC14})$$

Here W_{arch} is Connes's $W_{\mathbb{R}}$; his notation W_{∞} in the subsequent positivity discussion has the opposite sign. These local distributions are quoted with their human attribution, not claimed as new. The integral converges at 1 because the bracket vanishes there to at least first order, and at infinity because the test terms have compact support and the subtracted term decays.

The place-support lattice has the following explicit valuation into this collection of local distributions:

$$\mathcal{W}_A(f) = \mathbf{1}_A(\infty)W_{\text{arch}}(f) + \sum_{p \in A} W_p(f). \quad (\text{FLC15})$$

There are finitely many nonzero summands for each f : if $\text{supp } f \subset [e^{-T}, e^T]$, they require $m \log p \leq T$. Its exact support law is

$$\mathcal{W}_{A \cup B} + \mathcal{W}_{A \cap B} = \mathcal{W}_A + \mathcal{W}_B. \quad (\text{FLC16})$$

This follows term by term from the indicator identity and does not identify meet with convolution. For a bounded weight c define $\mathcal{W}_c = c_{\infty}W_{\text{arch}} + \sum_p c_p W_p$ in the same way. For $g \in C_c^{\infty}(\mathbb{R}_+^{\times})$ let $g^*(x) = \overline{g(x^{-1})}$ and $f = g * g^*$, with Haar measure dx/x . Substitution $x = e^u$ gives the exact correlation

$$H_g(a) = f(e^a) = \int_{\mathbb{R}} g(e^u) \overline{g(e^{u-a})} du, \quad H_g(-a) = \overline{H_g(a)}. \quad (\text{FLC17})$$

Consequently the entire changed prime term is

$$\boxed{\mathcal{W}_c(g * g^*) - c_{\infty}\mathcal{W}_1(g * g^*) = 2 \sum_{p, m \geq 1} (c_p - c_{\infty})(\log p) p^{-m/2} \Re H_g(m \log p).} \quad (\text{FLC18})$$

No archimedean term is omitted: it cancels because its two coefficients are c_{∞} . Every finite term is displayed. Combining FLC13 at $a = p$ gives the exact relation to the foundational support defect

$$\mathcal{W}_c(g * g^*) - c_{\infty}\mathcal{W}_1(g * g^*) = -2 \sum_{p, m \geq 1} \mathcal{D}(c)(p) p^{-m/2} \Re H_g(m \log p). \quad (\text{FLC19})$$

This is an exact morphism from the existing prime-support defect to the change in this defined place-weighted arithmetic functional. It does not assert that the choice of this functional is forced by an analytic trace of the full lattice geometry. The Weil criterion concerns the full-place distribution, with its stated Mellin vanishing conditions. Formula FLC19 calculates the change produced by each place-support weight before such a criterion can be applied to a restricted or mixed expression. In the convention where the desired Weil quadratic form is nonnegative, put $\mathcal{Q}_c(g) = -\mathcal{W}_c(g * g^*)$. Its defect is the negative of FLC19, so the coefficient is +2. The classical criterion in the quoted source is $\mathcal{Q}_1(g) \geq 0$ for all its compact tests with $\hat{g}(\pm i/2) = 0$; no such sign is assumed here.

6.4 The exact fate of the mixed branch kernel in this map

For a finite set I of labelled places, extend the place valuation linearly from support states by

$$\mathcal{V} : \mathbb{C}[\mathcal{P}(I)] \longrightarrow \mathcal{D}'(\mathbb{R}_+^{\times}), \quad \delta_A \longmapsto \sum_{v \in A} W_v, \quad (\text{FLC20})$$

where the archimedean label means W_{arch} . Its exact factorization is

$$\mathcal{V}\left(\sum_A c_A \delta_A\right) = \sum_{v \in I} \left(\sum_{A \ni v} c_A\right) W_v. \quad (\text{FLC21})$$

This follows by exchanging two finite sums. Hence $\ker \mathcal{B} \subseteq \ker \mathcal{V}$: every higher mixed vector FLC9 with $|T| \geq 2$ is killed by this specific local-distribution map. In particular the four-state vector FLC10 has image zero, since $\mathcal{W}_{\emptyset} - \mathcal{W}_{\{1\}} - \mathcal{W}_{\{2\}} + \mathcal{W}_{\{1,2\}} = 0$. The same finite-sum argument

proves that the foundational defect map $\delta_A \mapsto \Delta(A)$ kills this mixed kernel as well. Thus FLC19 calculates first place moments exactly. It does not silently calculate the higher joint moments in FLC11. The vector h has zero image under FLC20 even though its invariant line has the trace described after FLC10. These are two different specified maps: evaluating a distribution on a vector is not taking the trace on its invariant span. Their proved relationship is the kernel factorization FLC21. A trace realization retaining joint support must carry that kernel explicitly.

7 A compact Weil-admissible test detecting each prime-support defect

Sources and fixed conventions. The support defect is the original place-support construction of The Clankers, *Split Support Geometry*, Version 11, `def:adelic-support-defect`. Its rational-coordinate calculation is FLC12–13 in the present continuation. The local explicit-formula contributions and test-function convention are Alain Connes’s original TeX, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022](https://arxiv.org/abs/2602.04022), section `sectweilpos`, equations `wp` and `arch`, together with the immediately following statement of André Weil’s criterion. The exact formula below uses $W_{\text{arch}} = W_{\mathbb{R}}$ from that source; it does not use the opposite-sign quantity which the source subsequently names W_{∞} .

Let $\Omega = \{\infty\} \sqcup \mathbb{P}$, let $c \in \ell^{\infty}(\Omega; \mathbb{R})$, and retain

$$d_p = \mathcal{D}(c)(p) = (c_{\infty} - c_p) \log p, \quad \mathcal{W}_c = c_{\infty} W_{\text{arch}} + \sum_q c_q W_q, \quad Q_c(g) = -\mathcal{W}_c(g * g^*), \quad (\text{PDD1})$$

where

$$g^*(x) = \overline{g(x^{-1})}, \quad (g * g^*)(x) = \int_0^{\infty} g(y) \overline{g(y/x)} \frac{dy}{y}, \quad W_q(f) = (\log q) \sum_{m \geq 1} q^{-m/2} (f(q^m) + f(q^{-m})). \quad (\text{PDD2})$$

All these prime sums are finite for each compactly supported test function. The admissibility conditions in this convention are

$$\widehat{g}(i/2) = \int_0^{\infty} g(x) x^{1/2} \frac{dx}{x} = 0, \quad \widehat{g}(-i/2) = \int_0^{\infty} g(x) x^{-1/2} \frac{dx}{x} = 0. \quad (\text{PDD3})$$

Weil’s nonnegative quadratic form is Q_1 on these tests. The following computations prove the exact support correction; they do not assume a sign for Q_1 .

7.1 The completely specified compact test

Fix any rational prime p and set

$$a_p = \log p, \quad \varepsilon_p = \frac{1}{8} \log \frac{p+1}{p},$$

$$u_p(t) = \begin{cases} \exp\left(-\frac{1}{1 - (t/\varepsilon_p)^2}\right), & |t| < \varepsilon_p, \\ 0, & |t| \geq \varepsilon_p, \end{cases} \quad v_p(t) = u_p''(t) - \frac{1}{4} u_p(t). \quad (\text{PDD4})$$

The function u_p is real, smooth, nonzero and compactly supported. For smoothness at each endpoint, every one-sided derivative is a finite sum of powers of $[1 - (t/\varepsilon_p)^2]^{-1}$ and polynomials in t times the exponential in PDD4. The exponential tends to zero faster than each such power, so every derivative extends with value zero. At $|t| < \varepsilon_p$, writing $b(t) = 1 - (t/\varepsilon_p)^2$ gives the explicit original derivative and coefficient

$$v_p(t) = u_p(t) \left(\frac{4t^2}{\varepsilon_p^4 b(t)^4} - \frac{8t^2}{\varepsilon_p^4 b(t)^3} - \frac{2}{\varepsilon_p^2 b(t)^2} - \frac{1}{4} \right). \quad (\text{PDD5})$$

Both v_p and every derivative vanish at the support endpoints. Its nonzero amplitude is proved without an assumption about a numerical integral: integration by parts gives

$$\int_{\mathbb{R}} v_p(t) u_p(t) dt = - \int_{\mathbb{R}} |u'_p(t)|^2 dt - \frac{1}{4} \int_{\mathbb{R}} |u_p(t)|^2 dt < 0. \quad N_p := \int_{-\varepsilon_p}^{\varepsilon_p} |v_p(t)|^2 dt > 0. \quad (\text{PDD6})$$

The rightmost strict inequality follows because v_p cannot be the zero function in view of the preceding strictly negative integral. N_p is kept as this exact integral of the explicitly specified function; no change of its scale has been made.

For $\sigma \in \{1, -1\}$ define the exact multiplicative test function

$$g_{p,\sigma}(x) = v_p(\log x) + \sigma v_p(\log x - a_p), \quad f_{p,\sigma} = g_{p,\sigma} * g_{p,\sigma}^*. \quad (\text{PDD7})$$

It is real, smooth, and supported in the union $[e^{-\varepsilon_p}, e^{\varepsilon_p}] \cup [pe^{-\varepsilon_p}, pe^{\varepsilon_p}]$. For either $b = 1/2$ or $b = -1/2$, two integrations by parts with the vanishing endpoint derivatives give

$$\int_{\mathbb{R}} v_p(t) e^{bt} dt = (b^2 - 1/4) \int_{\mathbb{R}} u_p(t) e^{bt} dt = 0. \quad (\text{PDD8})$$

The integral for $v_p(t - a_p)$ is e^{ba_p} times this zero. Substitution $x = e^t$ now proves both conditions PDD3 for every $g_{p,\sigma}$.

7.2 Exact isolation from every other prime power

Put $C_p(b) = \int_{\mathbb{R}} v_p(t) v_p(t - b) dt$. It is real and even, $C_p(0) = N_p$, and $C_p(b) = 0$ when $|b| > 2\varepsilon_p$, because the two compact supports then do not meet. Expansion of PDD7 gives the full correlation, including both mixed terms:

$$f_{p,\sigma}(e^b) = 2C_p(b) + \sigma C_p(b - a_p) + \sigma C_p(b + a_p). \quad (\text{PDD9})$$

There are no suppressed phases or translated pieces in this identity.

For any integer $n \geq 2$ with $n \neq p$,

$$|\log n - a_p| \geq \log \frac{p+1}{p} = 8\varepsilon_p. \quad (\text{PDD10})$$

For $n \geq p+1$ this is monotonicity of the logarithm. For $n \leq p-1$ it follows from $p/(p-1) > (p+1)/p$, which is equivalent to $p^2 > p^2 - 1$; when $p = 2$ there is no such integer $n \geq 2$ on the lower side. Also $2\varepsilon_p < \log 2 \leq \log n$ and $2\varepsilon_p < a_p$ because $(p+1)/p \leq 3/2 < 2^4$. Formula PDD9 therefore implies

$$f_{p,\sigma}(q^m) = f_{p,\sigma}(q^{-m}) = \begin{cases} \sigma N_p, & q = p, m = 1, \\ 0, & \text{every other prime } q \text{ and integer } m \geq 1. \end{cases} \quad (\text{PDD11})$$

Indeed the $2C_p(\log(q^m))$ term vanishes by $\log(q^m) \geq \log 2$; the positive-shift argument $\log(q^m) + a_p$ is also outside the support. The remaining argument $\log(q^m) - a_p$ is outside it by PDD10 unless $q^m = p$. Unique prime factorization gives $q = p, m = 1$ in the latter case. Its contribution is $\sigma C_p(0)$. Evenness proves the reciprocal evaluation as well.

7.3 Detection, inversion, and the exact sign

Every local prime contribution is now calculated:

$$W_q(f_{p,\sigma}) = \begin{cases} 2\sigma(\log p)p^{-1/2}N_p, & q = p, \\ 0, & q \neq p. \end{cases} \quad (\text{PDD12})$$

The archimedean contribution cancels in the difference with coefficient c_∞ . Thus

$$\begin{aligned}\mathcal{W}_c(f_{p,\sigma}) - c_\infty \mathcal{W}_1(f_{p,\sigma}) &= -2\sigma d_p p^{-1/2} N_p, \\ Q_c(g_{p,\sigma}) - c_\infty Q_1(g_{p,\sigma}) &= 2\sigma d_p p^{-1/2} N_p.\end{aligned}\tag{PDD13}$$

These identities follow either directly from PDD12 or from FLC19 with the positive Weil-form sign accounted for. They include the archimedean term through its exact cancellation, rather than omitting it from a formula for Q_1 .

In particular, the map from the original support defect $\mathcal{D}(c)$ to its quadratic-form correction on Weil-admissible compact tests is injective. Each prime coordinate has the explicit recovery formula

$$d_p = \frac{\sqrt{p}}{2N_p} (Q_c(g_{p,1}) - c_\infty Q_1(g_{p,1})).\tag{PDD14}$$

If all corrections vanish, every d_p is zero. Conversely their vanishing makes the correction identically zero by FLC19. For rational $a \neq 0$ the remaining defect is then recovered as the original finite sum $\mathcal{D}(c)(a) = \sum_p \nu_p(a) d_p$. This is an exact inverse on the image of the defect map, not just a necessary bound.

There is also a definite sign calculation for the archimedean-free subspace. If $c_\infty = 0$ and $c_p \neq 0$ for some prime, PDD13 gives

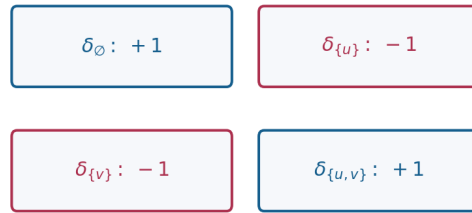
$$Q_c(g_{p,\sigma}) = -2\sigma c_p (\log p) p^{-1/2} N_p.\tag{PDD15}$$

Choosing $\sigma = \text{sgn}(c_p)$ makes this strictly negative, and the opposite sign makes it strictly positive. Consequently no nonzero real weight with $c_\infty = 0$ gives a nonnegative quadratic form on all the admissible tests. For a nonempty place support A omitting ∞ , take any $p \in A$: its indicator weight has the explicit negative value $Q_{1_A}(g_{p,1}) = -2(\log p) p^{-1/2} N_p$. This is a counterexample to positivity of that restricted place functional, not a counterexample to RH or an evaluation of the sign of the full-place form.

The higher mixed-support kernel is still retained. The injectivity in PDD14 starts with the place defect $\mathcal{D}(c)$, not with arbitrary formal linear combinations of support states. In a finite Boolean channel model, a formal vector $\sum_A t_A \delta_A$ maps to the place weights $c_i = \sum_{A \ni i} t_A$. Hence all the higher mixed vectors in FLC7–10 map to zero under both the defect and the local-place functional. For example $\delta_\emptyset - \delta_{\{1\}} - \delta_{\{2\}} + \delta_{\{1,2\}}$ has both place weights zero. The exact mixed kernel and its nonzero formal fixed trace are therefore not detected by PDD14. The maps in FLC20–21 retain that kernel explicitly; detecting every prime coordinate of their image does not erase it.

Mixed support recovered through its joint observation and exact algebra

The retained four-state vector



$$J_{\emptyset}(h) = J_u(h) = J_v(h) = 0$$

$$J_{\{u,v\}}(h) = 1$$

Marginals lose the vector.
The joint coefficient retains it.

Exact joint distribution

$$\mathcal{V}_2(h) = W_u \otimes W_v + W_v \otimes W_u$$

Coefficient matrix in the labelled basis

	W_u	W_v
W_u	0	1
W_v	1	0

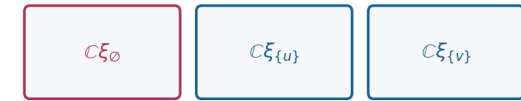
$$\mathcal{V}_2(h)(\psi_u \otimes \psi_v) = b_u b_v > 0$$

Separate variables; no product of distributions
at the same point. Decoder constants are retained.

Full additive support jet

$$\mathbb{C}[\eta]/(\eta^2)$$

Original action: $d_0 \Pi + d_1 R$



$$E_{\emptyset} + E_{\{u\}} + E_{\{v\}} = I - \Pi$$

$$\xi_{\emptyset} = h$$

The old scalar action specifies the sum.
Both individual branches miss this mixed factor.

Proofs: FLC10–11; FSJ2–8; JMD1–14. Local distributions: Alain Connes; full support base: Split Support Geometry v11.

Figure 2: The literal two-place vector h is invisible to the individual marginals. Its joint distribution has the two displayed cross terms, proved in JMD6–11; its algebra factor is FSJ4 and JMD20. The matrix contains tensor coefficients, not an assertion of Weil positivity. In fact JMD15 proves the joint bilinear functional indefinite. All decoder constants b_u, b_v are retained.

8 Exact recovery of joint support by local distributions

Sources and the constructed maps. The ring-null support lattice is the object in The Clankers, *Split Support Geometry*, Version 11, `def:lattice-split` and `thm:lattice-normal-form`. The source's adelic support valuation and its linear extension are `def:adelic-support-defect` and `thm:adelic-defect-linearization`. The local distributions used here are Alain Connes's *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022](https://arxiv.org/abs/2602.04022), section `sectweilpos`, equations `wp` and `arch`; the retained original author TeX has these at lines 854–867. We use $W_{\text{arch}} = W_{\mathbb{R}}$ there, not the opposite-sign quantity later named W_{∞} . The local distributions, their human attribution, and the original support states are retained without alteration.

The map $\delta_A \mapsto \sum_{v \in A} W_v$ is the specified distribution-valued place valuation in FLC20. This section constructs its joint observations and their exact relation to traces on the formal zero-support representation. The displayed insertion of the Connes distributions is part of the definition of the observable; it is not assumed to be an analytic trace forced by the base.

Let $G = \mathbb{R}_+^{\times}$ with Haar measure dx/x . Let I be a finite set of $n \geq 1$ distinct labelled places of $\mathbb{Q}$, and put $V = \mathbb{C}[\mathcal{P}(I)]$, with its original basis δ_A , $A \subseteq I$. For a vector $c = \sum_A c_A \delta_A$ define $J_T(c) = \sum_{A \supseteq T} c_A$ for every $T \subseteq I$. Here $J_{\emptyset}(c)$ is its total coefficient sum. The simultaneous individual-branch map has kernel $K = \{c : J_{\emptyset}(c) = 0, J_{\{i\}}(c) = 0 \text{ for all } i \in I\}$, as computed in FLC6–FLC10.

8.1 The exact joint coordinates of the mixed kernel

For each $S \subseteq I$, retain the vector $v_S = \sum_{A \subseteq I} (-1)^{|A \cap S|} \delta_A$. Its joint coordinates are

$$J_T(v_S) = \begin{cases} (-1)^{|S|} 2^{n-|T|}, & S \subseteq T, \\ 0, & S \not\subseteq T. \end{cases} \quad (\text{JMD1})$$

To prove this, write $A = T \cup B$ with $B \subseteq I \setminus T$. If $S \setminus T$ is nonempty, toggle any coordinate in that set; the paired signs cancel. If $S \subseteq T$, the sign is constantly $(-1)^{|S|}$ over all $2^{n-|T|}$ choices of B . The v_S form a basis: their squared norms in the original orthonormal support-state basis are 2^n , and for $S \neq U$ the inner product vanishes by toggling a coordinate in their symmetric difference. In particular the v_S with $|S| \geq 2$ are a basis of K . Indeed they belong to K by JMD1, and the conditions defining K have rank $n+1$: a linear relation among the functions 1 and $\mathbf{1}_{i \in A}$ vanishes first on $A = \emptyset$ and then on every singleton only when all its coefficients vanish. Thus $\dim K = 2^n - n - 1$, exactly the number of those basis vectors.

8.2 Literal compact tests for each local distribution

Put $\eta = (\log 2)/16$ and define

$$w_{\eta}(t) = \begin{cases} \exp\left(-\frac{1}{1 - (t/\eta)^2}\right), & |t| < \eta, \\ 0, & |t| \geq \eta, \end{cases} \quad (\text{JMD2})$$

$$\psi_{\infty}(e^t) = w_{\eta}(t - 4\eta) + w_{\eta}(t + 4\eta).$$

Every one-sided derivative of the bump is a polynomial in t and inverse powers of $1 - (t/\eta)^2$, multiplied by the displayed exponential. At either endpoint the exponential tends to zero faster than each such inverse power. Therefore w_{η} , and hence ψ_{∞} , is smooth with compact support. The latter is real, nonnegative and invariant under $x \mapsto x^{-1}$. Its logarithmic support satisfies $3\eta \leq |t| \leq 5\eta < \log 2$, and it vanishes at $x = 1$. Every $W_p(\psi_{\infty})$ is consequently zero. The exact archimedean value is

$$\beta := W_{\text{arch}}(\psi_{\infty}) = 2 \int_0^{\infty} \psi_{\infty}(e^t) \frac{e^{-t/2}}{1 - e^{-2t}} dt > 0. \quad (\text{JMD3})$$

This follows from FLC14, with its constant term and subtraction both zero because $\psi_\infty(1) = 0$. The integral is finite since its support avoids zero and infinity. It is strictly positive since its integrand is positive for $3\eta < t < 5\eta$.

For each rational prime p retain the exact PDD4–PDD7 functions: $a_p = \log p$, $\varepsilon_p = \frac{1}{8} \log((p+1)/p)$, $u_p(t) = \exp(-1/(1 - (t/\varepsilon_p)^2))$ for $|t| < \varepsilon_p$ and zero elsewhere, $v_p = u_p'' - u_p/4$, $g_{p,1}(e^t) = v_p(t) + v_p(t - a_p)$, and $f_{p,1} = g_{p,1} * g_{p,1}^*$. Keep $N_p = \int |v_p(t)|^2 dt > 0$ at its original scale. Define

$$\begin{aligned} \alpha_p &= 2(\log p)p^{-1/2}N_p > 0, & A_p &= W_{\text{arch}}(f_{p,1}), \\ \psi_p &= \beta f_{p,1} - A_p \psi_\infty, & b_\infty &= \beta, \quad b_p = \beta \alpha_p. \end{aligned} \tag{JMD4}$$

All coefficients here are the displayed integrals and factors; none of the original functions or support states has had its scale changed. The new function ψ_p is a specified linear combination used for decoding distributions.

For completeness the prime evaluations used in this construction follow directly from the original bump. Integration by parts gives $\int v_p u_p = -\int |u_p'|^2 - \frac{1}{4} \int u_p^2 < 0$, proving $N_p > 0$. Let $C_p(t) = \int v_p(s)v_p(s-t)ds$. It vanishes outside $[-2\varepsilon_p, 2\varepsilon_p]$, is even, and has value N_p at zero. Expanding the convolution yields $f_{p,1}(e^t) = 2C_p(t) + C_p(t - a_p) + C_p(t + a_p)$. For every integer $m \geq 2$, $m \neq p$, $|\log m - \log p| \geq \log((p+1)/p) = 8\varepsilon_p$. For $m \geq p+1$ this is monotonicity of the logarithm; for $m \leq p-1$ it follows from $p/(p-1) > (p+1)/p$. Also $2\varepsilon_p < \log 2$. Thus among all prime powers q^k , $k \geq 1$, only $q^k = p$ yields a nonzero evaluation, and that evaluation and its reciprocal both equal N_p . Therefore $W_q(f_{p,1}) = \alpha_p \mathbf{1}_{q=p}$. Together with JMD3 this proves the full duality

$$\boxed{W_v(\psi_u) = b_u \mathbf{1}_{v=u}, \quad b_u > 0,} \quad v, u \in \{\infty\} \sqcup \mathbb{P}, \tag{JMD5}$$

where the label ∞ on W_v means W_{arch} . For $u = p$, its archimedean value is $\beta A_p - A_p \beta = 0$; its only nonzero finite value is $\beta \alpha_p$ at p . For $u = \infty$, the assertions are JMD3 and the support calculation following JMD2.

These decoding functions are not asserted to be convolution squares or to satisfy Weil's two vanishing conditions. The original $g_{p,1}$ does satisfy them: for $b = \pm 1/2$, two integrations by parts give $\int v_p(t)e^{bt}dt = (b^2 - 1/4) \int u_p(t)e^{bt}dt = 0$, and translation by a_p multiplies this zero by e^{ba_p} . The distinction preserves the domains of both constructions.

8.3 The original support diagonal and the complete recovery

Retain $\mathcal{V} : V \rightarrow \mathcal{D}'(G)$ from FLC20, $\mathcal{V}\delta_A = \sum_{v \in A} W_v$. For $r \geq 1$ define the linearized set diagonal and its observation

$$\Delta^{(r)} : V \longrightarrow V^{\otimes r}, \quad \Delta^{(r)}\delta_A = \delta_A^{\otimes r}, \quad \mathcal{V}_r = \mathcal{V}^{\otimes r} \circ \Delta^{(r)}. \tag{JMD6}$$

The set map copies each original support state before its observations are taken; its extension is linear. In particular

$$\mathcal{V}_r(c) = \sum_{A \subseteq I} c_A \left(\sum_{v \in A} W_v \right)^{\otimes r}. \tag{JMD7}$$

Each term is a tensor product of distributions in separate variables, and the sum is finite, so the target is $\mathcal{D}'(G^r)$. Equivalently the expression acts as a continuous multilinear functional on r compact smooth functions. No product of singular distributions in one variable is used. The expression is linear in c ; it is not $\mathcal{V}(c)^{\otimes r}$, which has products of different coefficients c_{ACB} and loses the stated diagonal construction.

Given a tuple $(i_1, \dots, i_r)$ of labels from I , let T be its set of distinct entries. JMD5 gives the exact joint decoder

$$\boxed{\mathcal{V}_r(c)(\psi_{i_1} \otimes \dots \otimes \psi_{i_r}) = \left(\prod_{j=1}^r b_{i_j} \right) J_T(c).} \tag{JMD8}$$

Indeed the contribution of δ_A is $\prod_j \sum_{v \in A} W_v(\psi_{i_j}) = \prod_j b_{i_j} \mathbf{1}_{i_j \in A}$. The product of indicators is exactly $\mathbf{1}_{T \subseteq A}$. Summing with the original coefficients c_A proves the identity. Every b_{i_j} is strictly positive, so division by the displayed literal factor recovers the joint coordinate without an ambiguity.

Put $\mathcal{V}_0(c) = J_\emptyset(c)$. For every $0 \leq r \leq n$ the full observation through degree r has exact kernel

$$\boxed{\bigcap_{s=0}^r \ker \mathcal{V}_s = \{c : J_T(c) = 0 \text{ for all } |T| \leq r\} = \text{span}\{v_S : |S| > r\}. \quad (\text{JMD9})}$$

To prove the first equality, JMD8 recovers every nonempty coordinate of degree at most r , and $\mathcal{V}_0$ gives the empty coordinate. Conversely expanding JMD7 gives coefficients $J_T(c)$ with $|T| \leq s$, so their vanishing makes each distribution zero. The stated Walsh vectors lie in the kernel by JMD1. For dimension and complete recovery, the full moment map has inverse

$$\boxed{c_A = \sum_{T \supseteq A} (-1)^{|T \setminus A|} J_T(c). \quad (\text{JMD10})}$$

Substitute $J_T(c) = \sum_{B \supseteq T} c_B$. The coefficient at c_A is one. For $B \supsetneq A$, the coefficient is $\sum_{A \subseteq T \subseteq B} (-1)^{|T \setminus A|} = 0$ by pairing subsets differing in one element of $B \setminus A$. This proves the inverse. Consequently all the functions $A \mapsto \mathbf{1}_{T \subseteq A}$ are independent, and the subfamily $|T| \leq r$ has rank $\sum_{k=0}^r \binom{n}{k}$. The kernel dimension is $\sum_{k=r+1}^n \binom{n}{k}$, exactly the number of displayed independent Walsh vectors, completing JMD9. Thus degrees zero through n recover every coefficient. On the mixed kernel K , the degrees zero and one already vanish; precisely the remaining joint degrees two through n must be retained, and JMD8–JMD10 recover them all.

For two distinct places u, v , put $\mathbf{h} = \delta_\emptyset - \delta_{\{u\}} - \delta_{\{v\}} + \delta_{\{u,v\}}$ in their four-state support space. Then $\mathcal{V}\mathbf{h} = 0$, but the full second observation is

$$\boxed{\begin{aligned} \mathcal{V}_2(\mathbf{h}) &= W_u \otimes W_v + W_v \otimes W_u, \\ \mathcal{V}_2(\mathbf{h})(\psi_u \otimes \psi_v) &= b_u b_v > 0. \end{aligned} \quad (\text{JMD11})}$$

Expanding $(W_u + W_v)^{\otimes 2}$ cancels exactly the two unmixed terms subtracted by the singleton states. JMD5 evaluates the two remaining terms as $b_u b_v$ and zero, respectively. This also proves that the group-like diagonal does not descend through the quotient retaining only individual branch observations. Indeed $\mathbf{h}$ is in the branch kernel while $(\mathcal{V} \otimes \mathcal{V})\Delta^{(2)}\mathbf{h} \neq 0$; the map $\mathcal{V}$ itself factors through that quotient. If its kernel were a coideal, the latter tensor observation would vanish. The nonzero tensor in JMD11 is the exact defect of that descent.

8.4 The corresponding finite-dimensional trace

Define operators on the original support-state basis by

$$E_i \delta_A = \mathbf{1}_{i \in A} \delta_A, \quad E_T = \prod_{i \in T} E_i, \quad D_c \delta_A = c_A \delta_A. \quad (\text{JMD12})$$

The E_i are commuting projections and $E_\emptyset = I_V$. These are diagonal observables on the formal support-state space, not an identification with an additive-monoid algebra. The idele action fixes every such zero state by FLC1: $(u, 1_L)(0, A) = (0, A)$. Hence on this finite-dimensional representation its integrated operator for an integrable h_0 on the idele class group C is $U(h_0) = (\int_C h_0(u) d^\times u) I_V$. Taking the trace by summing the diagonal entries proves

$$\boxed{\text{Tr}_V(D_c E_T U(h_0)) = J_T(c) \int_C h_0(u) d^\times u. \quad (\text{JMD13})}$$

Thus the joint moments have an exact finite-dimensional trace realization on the actual formal zero-support representation.

Define the specified distribution-valued diagonal observable $A(f) = \sum_{i \in I} W_i(f) E_i$. Multiplying its diagonal entries proves the complete connecting map

$$\boxed{\text{Tr}_V(D_c A(f_1) \cdots A(f_r)) = \mathcal{V}_r(c)(f_1 \otimes \cdots \otimes f_r).} \quad (\text{JMD14})$$

In detail, the entry at δ_A of the product is $c_A \prod_j \sum_{i \in A} W_i(f_j)$. Summation is JMD7 evaluated on the tensor of the f_j . Insertion of $U(h_0)$ multiplies this exact formula by $\int_C h_0(u) d^\times u$.

For the four-state vector $\mathbf{h}$, the trace on its invariant line is $\int_C h_0$, as in FLC10. The different operator $D_{\mathbf{h}}$ on V has integrated trace zero because $J_{\emptyset}(\mathbf{h}) = 0$. The insertion of $E_u E_v$ changes that trace to $\text{Tr}_V(D_{\mathbf{h}} E_u E_v U(h_0)) = \int_C h_0$, since $J_{\{u,v\}}(\mathbf{h}) = 1$. The traced operators and spaces are therefore specified and their values agree with the complete joint reconstruction.

JMD13 uses the representation forced by the stated action on the formal zero-support states. JMD14 additionally inserts the Connes local distributions by the displayed definition of $A(f)$. No equality between that insertion and an analytic cohomological trace of the original programme has been proved here. The exact maps obtained so far are the support diagonal, its tensor-valued local observations, the full finite joint inverse, and their stated finite-dimensional trace realization.

8.5 The exact sign of the joint two-place bilinear form

For the two-place vector $\mathbf{h}$ in JMD11 keep the literal test combinations $\varphi_+ = \psi_u + \psi_v$ and $\varphi_- = \psi_u - \psi_v$. JMD5 gives $W_u(\varphi_{\pm}) = b_u$ and $W_v(\varphi_{\pm}) = \pm b_v$. Evaluation of both tensor summands in JMD11 therefore proves

$$\boxed{\mathcal{V}_2(\mathbf{h})(\varphi_{\pm} \otimes \varphi_{\pm}) = \pm 2b_u b_v.} \quad (\text{JMD15})$$

Since both constants are strictly positive, the real symmetric bilinear functional $(f, g) \mapsto \mathcal{V}_2(\mathbf{h})(f \otimes g)$ is indefinite. This sign calculation uses the displayed joint functional and decoding tests. It is not a counterexample to Weil positivity: these decoders were not asserted to be admissible convolution squares, and the bilinear distribution in JMD15 is not the full-place Weil quadratic form.

8.6 Exact relation to support addition and its incidence algebra

Let $L = \mathcal{P}(I)$ and retain the zero-support elements $z_A = (0, A)$ in $G_L(\mathbb{Z})$. Their addition is $z_A + z_B = z_{A \cup B}$, directly from the original coefficient definition. Thus their submonoid gives a subalgebra $\mathbb{C}[L, \cup] \hookrightarrow \mathbb{C}[G_L(\mathbb{Z}), +]$, whose unit is $[z_{\emptyset}]$. The inclusion is injective because its distinct basis elements remain distinct basis elements of the larger monoid algebra. Its left regular action on the original zero-state space V is

$$S_B \delta_A = \delta_{A \cup B}, \quad S_B S_C = S_{B \cup C}, \quad S_{\emptyset} = I_V. \quad (\text{JMD16})$$

Composition proves the product identity on every basis vector. The representation is faithful: applying $\sum_B t_B S_B$ to $\delta_{\emptyset}$ yields $\sum_B t_B \delta_B$, so a zero operator forces all coefficients to vanish. This is the restriction of support addition, not an asserted restriction of every element of the larger monoid algebra to V ; addition by a nonzero amplitude can leave the zero-state sector.

The diagonal place observations in JMD12 and the support-addition operators have the exact commutator

$$\boxed{E_i S_B - S_B E_i = \mathbf{1}_{i \in B} S_B (I_V - E_i).} \quad (\text{JMD17})$$

Indeed the left side applied to δ_A is $(\mathbf{1}_{i \in A \cup B} - \mathbf{1}_{i \in A}) \delta_{A \cup B}$. The difference of indicators is $\mathbf{1}_{i \in B}(1 - \mathbf{1}_{i \in A})$, which is the right side. In particular $E_i S_{\{i\}} = S_{\{i\}}$ and $S_{\{i\}} E_i = E_i$; their

difference is nonzero on $\delta_\emptyset$. Therefore the E_i are not being identified with elements of the commutative regular image of $\mathbb{C}[L, \cup]$. Their exact relation is JMD17.

Let $\mathfrak{J}$ be the subalgebra of $\text{End}_{\mathbb{C}}(V)$ generated by all S_B and all E_i . For $A \subseteq I$ put $F_A = \prod_{i \in A} E_i \prod_{i \notin A} (I_V - E_i)$. It is the projection onto the literal line $\mathbb{C}\delta_A$: its value at δ_C is the product of indicators expressing $C = A$. For $A \subseteq B$ define $M_{B,A}\delta_C = \mathbf{1}_{C=A}\delta_B$. Then the generated algebra is exactly

$$\boxed{\mathfrak{J} = \text{span}\{M_{B,A} : A \subseteq B \subseteq I\}, \quad M_{B,A} = F_B S_B F_A, \quad \dim \mathfrak{J} = 3^n.} \quad (\text{JMD18})$$

For the displayed product, F_A kills every basis vector except δ_A ; S_B sends it to δ_B when $A \subseteq B$; and F_B fixes the result. Thus every displayed matrix unit belongs to $\mathfrak{J}$. Conversely these matrix units span a subalgebra, because $M_{B,A}M_{D,C} = \mathbf{1}_{A=D}M_{B,C}$ and the nonzero case has $C \subseteq A \subseteq B$. Each S_B sends δ_A to a basis vector labelled by a superset of A , so it lies in that subalgebra; the diagonal E_i lie there as well. This proves equality. The units are independent by their distinct matrix entries. There are 3^n pairs $A \subseteq B$: at each coordinate the choices are membership in neither set, in B only, or in both. This proves the dimension and identifies the generated algebra as the inclusion-incidence matrix algebra of the original Boolean lattice. In particular the point projections reconstruct every diagonal insertion in JMD13–JMD14 through $D_c = \sum_A c_A F_A$.

The exact coordinate change diagonalizing the support-addition subalgebra is also available. For $T \subseteq I$ put

$$w_T = \sum_{A \supseteq T} (-1)^{|A \setminus T|} \delta_A. \quad S_B w_T = \mathbf{1}_{B \subseteq T} w_T. \quad (\text{JMD19})$$

If $B \subseteq T$, every A in the sum already contains B , so S_B fixes every summand. Otherwise choose $i \in B \setminus T$. Pair the summands indexed by A and $A \cup \{i\}$ with $i \notin A$. Their signs are opposite and their images under S_B coincide; they cancel. This proves the eigenvalue formula. The transformation from the δ_A to the w_T is triangular with diagonal entries one when subsets are ordered by cardinality, and its inverse is $\delta_T = \sum_{A \supseteq T} w_A$. The inverse follows by the same alternating-subset sum as JMD10. Thus these are exact coordinates on the same support-state space.

For $I = \{u, v\}$, $w_\emptyset$ is exactly $\mathbf{h}$. As an element of the support-addition monoid algebra it is

$$\mathbf{h} = ([z_\emptyset] - [z_{\{u\}}])([z_\emptyset] - [z_{\{v\}}]), \quad \mathbf{h}^2 = \mathbf{h}, \quad S_B \mathbf{h} = 0 \quad (B \neq \emptyset). \quad (\text{JMD20})$$

The product expands to its original four coefficients because $[z_A][z_B] = [z_{A \cup B}]$. Each factor is idempotent, since $[z_{\{i\}}]^2 = [z_{\{i\}}]$, and they commute; their product is idempotent. The final identity is JMD19. Nevertheless its tensor-valued observation is the nonzero JMD11, and the diagonal insertion $E_u E_v$ extracts its coefficient one at $\delta_{\{u,v\}}$. These formulas connect the mixed vector to support addition, joint local-distribution observations, and diagonal trace insertions without identifying their different algebra structures.

8.7 The exact mixed sign at every even tensor order

For the same two-place vector $\mathbf{h}$, expansion of JMD7 gives, for every integer $r \geq 1$,

$$\mathcal{V}_r(\mathbf{h}) = (W_u + W_v)^{\otimes r} - W_u^{\otimes r} - W_v^{\otimes r}. \quad (\text{JMD21})$$

The empty support contributes the zero distribution; the other three states have precisely the coefficients $-1, -1, +1$ of $\mathbf{h}$. This proves JMD21 directly, including the zero result at $r = 1$. Evaluating at the literal decoding functions of JMD15 yields

$$\begin{aligned} \mathcal{V}_r(\mathbf{h})(\varphi_+^{\otimes r}) &= (b_u + b_v)^r - b_u^r - b_v^r > 0 \quad (r \geq 2), \\ \mathcal{V}_{2m}(\mathbf{h})(\varphi_-^{\otimes 2m}) &= (b_u - b_v)^{2m} - b_u^{2m} - b_v^{2m} < 0 \quad (m \geq 1). \end{aligned} \quad (\text{JMD22})$$

The first strict inequality follows from the positive mixed terms in the binomial expansion, since $b_u, b_v > 0$. For the second, $|b_u - b_v| < \max(b_u, b_v)$, so the first term is strictly smaller than even the larger of the two subtracted terms. Thus every even-order diagonal evaluation has both signs.

The negative evaluation has an exact nonvanishing root limit:

$$\lim_{m \rightarrow \infty} \left[b_u^{2m} + b_v^{2m} - (b_u - b_v)^{2m} \right]^{1/(2m)} = \max(b_u, b_v) > 0. \quad (\text{JMD23})$$

Put $B = \max(b_u, b_v)$ and $b = \min(b_u, b_v) > 0$. The quantity inside brackets is bounded above by $2B^{2m}$ and below by $B^{2m}[1 - (1 - b/B)^{2m}]$. The root of the upper factor 2 tends to one; the lower factor tends to one and has the same root limit, by taking its logarithm divided by $2m$. The squeeze proves the formula, also when $b = B$. This is a calculation for the constructed joint observation of the original mixed support vector. It does not identify that observation with Deligne's Euler coefficients or the full Weil form. It proves that an even tensor order alone does not provide positivity for this particular surviving mixed channel, and gives its complete size.

9 Closed prime entries for the original arithmetic test vectors

The original marked-product programme source is retained unchanged. Its retained filename is `TAU_ORIGINAL.tex` in the source directory. It defines

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad D = -x \frac{d}{dx}, \quad f_P = P(D)\phi_*, \quad (\text{GPE1})$$

for a polynomial $P(s) = \sum_{r=0}^d p_r s^r$. These exact vectors, including the two original coefficients and the sign of D , are used below. The calculation evaluates FLC19 on them without replacing them by a generic positive test function.

Put $b_1^{(0)} = -6\pi$, $b_2^{(0)} = 4\pi^2$ and all other $b_j^{(0)} = 0$. Set missing indices equal to zero. Direct differentiation gives the finite recurrence

$$b_j^{(r+1)} = -2j b_j^{(r)} + 2\pi b_{j-1}^{(r)}, \quad b_j = \sum_{r=0}^d p_r b_j^{(r)}, \quad f_P(x) = e^{-\pi x^2} \sum_{j=1}^{d+2} b_j x^{2j}. \quad (\text{GPE2})$$

Indeed $D(x^{2j} e^{-\pi x^2}) = (-2j x^{2j} + 2\pi x^{2j+2}) e^{-\pi x^2}$. Induction proves the recurrence at every order, retaining every coefficient, including cancellations to zero.

For the source's Mellin convention $\mathcal{M}f(s) = \int_0^\infty f(x) x^s dx/x$, direct integration with $t = \pi x^2$ gives, for $\Re s > -2$,

$$\mathcal{M}\phi_*(s) = 2\pi^{-s/2} \Gamma(s/2 + 2) - 3\pi^{-s/2} \Gamma(s/2 + 1) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2). \quad (\text{GPE3})$$

The first equality follows term by term from the defining gamma integral; the second uses $\Gamma(z+1) = z\Gamma(z)$ and extends across removable points by continuity. At both endpoints of integration $x^s f_P(x)$ tends to zero for $\Re s > -2$. Integration by parts hence gives $\mathcal{M}(Df) = s\mathcal{M}f$ and

$$\mathcal{M}f_P(s) = P(s)\Phi(s), \quad \Phi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2), \quad \mathcal{M}f_P(1) = 0, \quad \mathcal{M}f_P(0) = -P(0). \quad (\text{GPE4})$$

The exact passage to Connes's multiplicative test convention is $g_P(x) = x^{1/2} f_P(x)$; the factor is part of the map and is retained. Define its correlation, with the original multiplicative Haar measure, by $H_P(a) = (g_P * g_P^*)(e^a)$. Substitution as in FLC17 yields

$$H_P(a) = e^{-a/2} \int_0^\infty f_P(x) \overline{f_P(e^{-a}x)} dx. \quad (\text{GPE5})$$

The integral converges absolutely at both ends. Substitution of GPE2 and the gamma integral give the closed expression

$$H_P(a) = \frac{1}{2} \sum_{j,l=1}^{d+2} b_j \bar{b}_l \frac{\Gamma(j+l+1/2)}{\pi^{j+l+1/2}} \frac{e^{-(2l+1/2)a}}{(1+e^{-2a})^{j+l+1/2}}. \quad (\text{GPE6})$$

To verify each coefficient, the relevant integral is $\int_0^\infty x^{2j+2l} e^{-\pi(1+e^{-2a})x^2} dx = \frac{1}{2} \Gamma(j+l+1/2) [\pi(1+e^{-2a})]^{-j-l-1/2}$; its outside factors are exactly those shown. All sums are finite. Complex conjugation and translation in GPE5 prove $H_P(-a) = \overline{H_P(a)}$.

For $a \geq 0$, define the explicit nonnegative constant

$$K_P = \frac{1}{2} \sum_{j,l=1}^{d+2} |b_j \bar{b}_l| \frac{\Gamma(j+l+1/2)}{\pi^{j+l+1/2}}. \quad |H_P(a)| \leq K_P e^{-5a/2}. \quad (\text{GPE7})$$

The denominator in GPE6 is at least one, and $l \geq 1$ gives $2l+1/2 \geq 5/2$. The triangle inequality proves the bound with this literal constant. It applies to $P = 0$ as well. Consequently the support-dependent prime contribution is the absolutely convergent, explicitly evaluable series

$$\mathcal{P}_c(P) = 2 \sum_{p,m \geq 1} c_p(\log p) p^{-m/2} \Re H_P(m \log p), \quad (\text{GPE8})$$

with every entry given by GPE6. Truncation by the original prime-power integer $p^m \leq N$, $N \geq 2$ an integer, has proved error

$$\left| \mathcal{P}_c(P) - 2 \sum_{p^m \leq N} c_p(\log p) p^{-m/2} \Re H_P(m \log p) \right| \leq \|c\|_\infty K_P \frac{\log N + 1/2}{N^2}. \quad (\text{GPE9})$$

Indeed a prime-power term has absolute value at most $2\|c\|_\infty K_P (\log p) (p^m)^{-3}$. Grouping by $n = p^m$ and using $\log p \leq \log n$ bounds the tail by $2\|c\|_\infty K_P \sum_{n>N} \log n / n^3$. The function $\log x / x^3$ decreases for $x \geq 2$, so this is at most $2\|c\|_\infty K_P \int_N^\infty \log x / x^3 dx = \|c\|_\infty K_P (\log N + 1/2) / N^2$.

The archimedean functional in this same convention is

$$\begin{aligned} \mathcal{A}(P) &= (\log(4\pi) + \gamma) H_P(0) \\ &+ \int_0^\infty \frac{e^{-a/2} [H_P(a) + H_P(-a) - 2e^{-a/2} H_P(0)]}{1 - e^{-2a}} da. \end{aligned} \quad (\text{GPE10})$$

This is FLC14 after $x = e^a$, with no constants dropped. The numerator vanishes at $a = 0$ to at least first order, so the apparent singularity is integrable. At infinity GPE7 and its conjugate give an integrable bound; the subtracted term is $O(e^{-a})$. Thus $\mathcal{W}_c(P) := c_\infty \mathcal{A}(P) + \mathcal{P}_c(P)$ is a well-defined extension of those local expressions to these original noncompact test vectors. Its exact defect is

$$\mathcal{W}_c(P) - c_\infty \mathcal{W}_1(P) = -2 \sum_{p,m \geq 1} \mathcal{D}(c)(p) p^{-m/2} \Re H_P(m \log p). \quad (\text{GPE11})$$

The series converges by GPE7 because $|\mathcal{D}(c)(p)| \leq 2\|c\|_\infty \log p$. This gives the support defect on the actual programme vectors, not only on unspecified test functions. It makes no unproved assertion about its sign or an exchange with an infinite zero sum.

Finally, for the Fourier–Mellin convention in FLC14, $\hat{g}_P(t) = \mathcal{M}f_P(1/2 - it)$. In particular

$$\hat{g}_P(i/2) = 0, \quad \hat{g}_P(-i/2) = -P(0). \quad (\text{GPE12})$$

Thus both separate vanishing conditions hold precisely when $P(0) = 0$. For every P , the two cross-products $\hat{g}_P(i/2) \overline{\hat{g}_P(-i/2)}$ and its conjugate are zero. These are different assertions, and GPE12 retains both.

A The complete original amplitude memory calculation

The following derivation works in the original amplitude spaces and metric. Its finite sums and compositions lift by FLR4 and CLM19; its analytic evolution is recovered by CLM8–17. Its Deligne comparison does not assert an equivalence of the programme with Deligne’s geometric hypotheses. The exact kernel and tensor defects are retained. The reading coverage of the local Deligne transcription is stated at its use, separately from the original author sources of Connes and the original programme TeX.

B The original support action on prime powers, kernel memory and tensor amplification

This calculation uses the exact objects and metric of ST1–ST20 in *What the original observation retains of split support*, result SZ-20260922-028. It neither substitutes a new source metric nor discards the original observation kernel. The human mathematical precedents for the final comparison are Pierre Deligne, *La conjecture de Weil. II*, Publications Mathématiques de l’IHÉS **52** (1980), 137–252, §§1.4–1.5 and 3.2, and the Grothendieck trace formula he uses in 1.4.5.1. The supplied English TeX is a local translation/transcription, not an original author TeX source. It has been read at 1.4.3–1.5.2 and 3.2.1–3.2.13; its compressed presentation of §3.2 does not constitute verification of all intermediate assertions in Deligne’s original proof. Every finite-dimensional assertion made below is proved here.

B.1 The exact prime-power spectrum of the original source

Retain the original integer $k \geq 17$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$, the original simple quartet with $\delta, \gamma > 0$, the admitted period, and each literal cutoff $q-1 \leq N \leq 2q$. Set

$$\begin{aligned} \lambda_{ab} &= (2b-k)\gamma - i(2a-k)\delta, & Q_k(y) &= \prod_{a,b=0}^k (y - \lambda_{ab}), & E &= \mathbb{C}[y]/Q_k, & M[f] &= [yf], \\ s &= k/2 + iy, & F_p &= \exp(i(\log p)M), & \tilde{F}_p &= p^{k/2}F_p. \end{aligned} \tag{TDP1}$$

Here p is a rational prime. The map $s = k/2 + iy$ is the original centred Mellin coordinate map; the scalar $p^{k/2}$ is retained in the uncentred action. The parameters k and q have not been replaced by the number of factors in any later tensor construction.

The q roots in TDP1 are distinct: equality of two roots forces equality of their real and imaginary parts, and hence $a = a'$ and $b = b'$. The exact cardinal polynomials

$$\ell_{ab}(y) = \prod_{(c,d) \neq (a,b)} \frac{y - \lambda_{cd}}{\lambda_{ab} - \lambda_{cd}} \tag{TDP2}$$

therefore form a basis of E . Evaluation at the roots proves $M\ell_{ab} = \lambda_{ab}\ell_{ab}$ in E . Consequently, for every integer $m \geq 0$,

$$\begin{aligned} F_p^m \ell_{ab} &= p^{m(2a-k)\delta} \exp(im(2b-k)\gamma \log p) \ell_{ab}, \\ \tilde{F}_p^m \ell_{ab} &= p^{mk/2+m(2a-k)\delta} \exp(im(2b-k)\gamma \log p) \ell_{ab}. \end{aligned} \tag{TDP3}$$

This follows by applying the convergent exponential series to each cardinal vector. Different prime-action eigenvalues may coincide because of their phases; the labelled vectors and their multiplicities remain. In particular the exact largest eigenvalue modulus is $p^{mk\delta}$ for F_p^m and $p^{mk/2+mk\delta}$ for $\tilde{F}_p^m$. These are statements about the programme’s stipulated source, not identifications of a finite trace with the scalar prime evaluations in Weil’s explicit formula.

Let G_N be the complete original source metric, let $K = \ker \Lambda$, and retain the original section

$$\begin{aligned} Q_B &= (\Lambda G_N^{-1} \Lambda^*)^{-1}, & L &= G_N^{-1} \Lambda^* Q_B, & J &= L Q_B^{-1/2}, \\ J^\dagger J &= I_B, & P &= J J^\dagger = L \Lambda, & Q &= I_E - P, & \Phi(X) &= J^\dagger X J. \end{aligned} \quad (\text{TDP4})$$

The dagger uses G_N on E , its restriction on K , and the stated Euclidean coordinates on B . Let $\iota_K : K \hookrightarrow E$ be the isometric inclusion. The map $W : B \oplus K \rightarrow E$, $W(y, z) = Jy + \iota_K z$, is unitary for these metrics, since the two summands are orthogonal, $\text{im } J = K^\perp$, and their sum is E . Define the full blocks

$$W^\dagger M W = \begin{pmatrix} H & B_{01} \\ B_{10} & D_K \end{pmatrix}, \quad \begin{aligned} H &= J^\dagger M J, & B_{01} &= J^\dagger M \iota_K, \\ B_{10} &= \iota_K^\dagger M J, & D_K &= \iota_K^\dagger M \iota_K. \end{aligned} \quad (\text{TDP5})$$

No invariance of K under M is assumed or used.

B.2 The entire compression and the complete return through the kernel

For every $z \in \mathbb{C}$ put $U(z) = J^\dagger e^{zM} J$ and $Y(z) = \iota_K^\dagger e^{zM} J$. Differentiating the entire exponential series and applying TDP5 gives

$$\begin{aligned} U' &= H U + B_{01} Y, & Y' &= B_{10} U + D_K Y, & U(0) &= I_B, & Y(0) &= 0, \\ Y(z) &= \int_0^z e^{(z-t)D_K} B_{10} U(t) dt, \\ U'(z) &= H U(z) + \int_0^z \mathcal{K}(z-t) U(t) dt, & \mathcal{K}(t) &= B_{01} e^{tD_K} B_{10}. \end{aligned} \quad (\text{TDP6})$$

Integrals are along the straight segment, with the usual complex orientation. The second equation follows by differentiating $e^{-zD_K} Y(z)$ and integrating; substitution proves the third. Thus TDP6 is an identity of entire functions, including purely imaginary prime arguments. A second variation-of-constants calculation gives

$$U(z) - e^{zH} = \int_0^z \int_0^t e^{(z-t)H} B_{01} e^{(t-v)D_K} B_{10} U(v) dv dt. \quad (\text{TDP7})$$

The difference is the full original-kernel return, including its complex signs. No positive sign for this operator is asserted.

Insertion of $I_E = P + Q$ in $e^{(z+w)M}$ proves the exact product defect

$$\begin{aligned} U(z+w) - U(z)U(w) &= J^\dagger e^{zM} Q e^{wM} J, \\ U(mz) - U(z)^m &= \sum_{j=1}^{m-1} U(z)^{m-1-j} J^\dagger e^{zM} Q e^{jzM} J \quad (m \geq 1). \end{aligned} \quad (\text{TDP8})$$

For the second identity the sum is empty at $m = 1$. If the assertion is known for m , write $U((m+1)z) - U(z)^{m+1} = U(z)(U(mz) - U(z)^m) + U((m+1)z) - U(z)U(mz)$ and use the first identity. This proves it for all integers m . Specializing $z = i \log p$ gives the exact correction between the observed prime-power action and powers of the observed one-prime action. For the uncentred action both sides acquire precisely $p^{mk/2}$, with no change to any of the kernel factors.

The same defect determines the local determinant. Write $W^\dagger F_p W = \begin{pmatrix} A_p & B_p \\ C_p & D_p \end{pmatrix}$. Over formal power series in u , every matrix with constant term I is invertible. Block Gaussian elimination then gives

$$\begin{aligned} \sum_{m \geq 0} \Phi(F_p^m) u^m &= [I_B - u A_p - u^2 B_p (I_K - u D_p)^{-1} C_p]^{-1}, \\ \det(I_E - u F_p) &= \det(I_K - u D_p) \det(I_B - u A_p) \mathcal{E}_p(u), \\ \mathcal{E}_p(u) &= \det[I_B - u^2 (I_B - u A_p)^{-1} B_p (I_K - u D_p)^{-1} C_p]. \end{aligned} \quad (\text{TDP9})$$

To verify the first identity, expand $(I_E - uF_p)^{-1}$ as its geometric series and take its upper-left block. Eliminate the lower-left block using the invertible lower-right block $I_K - uD_p$; its Schur complement is the displayed inverse. Taking determinants proves the second identity, and factoring $I_B - uA_p$ proves the third. Since both sides are rational functions, the identities also hold wherever the displayed inverses exist, and extend across removable singularities after clearing denominators. They retain every pole cancellation carried by K .

The formal logarithmic derivative, or the identity $-\log \det(I - uT) = \sum_{m \geq 1} \text{tr}(T^m)u^m/m$, yields

$$-\log \mathcal{E}_p(u) = \sum_{m \geq 1} \frac{\text{tr}(F_p^m) - \text{tr}(A_p^m) - \text{tr}(D_p^m)}{m} u^m, \quad (\text{TDP10})$$

$$[u^2](-\log \mathcal{E}_p(u)) = \text{tr}(B_p C_p).$$

Indeed the diagonal blocks of F_p^2 have traces $\text{tr}(A_p^2) + \text{tr}(B_p C_p)$ and $\text{tr}(D_p^2) + \text{tr}(C_p B_p)$; cyclicity of the finite trace proves the coefficient claim. These traces are generally complex. Positivity of Φ on positive operators does not change this calculation into a nonnegative Euler coefficient.

B.3 The support algebra exhibits the cancellation explicitly

The original support representation and its boundary operator are

$$\begin{aligned} \rho(c_0, c_1 + c_2\eta) &= c_0(I - \Pi) + c_1\Pi + c_2R, & \Pi &= ee^\dagger + ff^\dagger, & R &= \epsilon f e^\dagger, \\ e^\dagger f &= 0, & \|e\| &= \|f\| = 1, & \epsilon > 0, & R^2 = 0, \\ [\tau] &\mapsto I_E, & [n] &\mapsto \Pi + nR. \end{aligned} \quad (\text{TDP11})$$

The symbol multiplication in the last line represents addition in the original monoid $\mathbb{Z} \sqcup \{\tau\}$. In particular $[p]^m = [mp]$ there, not $[p^m]$. The original τ action cannot be identified with the multiplicative prime-power law by replacing one of these indices. The exact one-parameter group in the same algebra is $\rho(1, 1 + z\eta) = I_E + zR = e^{zR}$; at integral $z = n$ this is the image of $[\tau] + [n] - [0]$. Its multiplication law follows from $R^2 = 0$.

Retain the measured columns and all their phases:

$$\begin{aligned} u &= J^\dagger e, & v &= J^\dagger f, & \begin{pmatrix} a & r \\ \bar{r} & d \end{pmatrix} &= [u, v]^* [u, v], & \alpha &= 1 - a, & \mathfrak{d} &= ad - |r|^2, \\ T &= \Phi(R) = \epsilon v u^*, & \kappa &= \epsilon r, & T^2 &= \kappa T, \\ e_K &= \iota_K^\dagger e, & f_K &= \iota_K^\dagger f, & \|e_K\|^2 &= \alpha, & \|f_K\|^2 &= 1 - d, & e_K^* f_K &= -r. \end{aligned} \quad (\text{TDP12})$$

The identity $T^2 = \kappa T$ is direct multiplication using $u^*v = r$. The last line follows by inserting $P + Q$ in the inner products of e, f . The support map preserves the original nilpotent on E , but compression can give it the nonzero eigenvalue κ . Since $\mathfrak{d} > 0$ in the original finite area guards, neither u nor v is zero, so $T \neq 0$.

The complete R blocks, its memory and its two exponentials are

$$\begin{aligned} W^\dagger R W &= \begin{pmatrix} T & \epsilon v e_K^* \\ \epsilon f_K u^* & \epsilon f_K e_K^* \end{pmatrix}, & \mathcal{K}_R(z) &= -\kappa e^{-\kappa z} T, \\ \Phi(e^{zR}) &= I_B + zT, & e^{zT} &= I_B + f_\kappa(z)T, & f_\kappa(z) &= \begin{cases} (e^{\kappa z} - 1)/\kappa, & \kappa \neq 0, \\ z, & \kappa = 0. \end{cases} \end{aligned} \quad (\text{TDP13})$$

For the memory identity, the lower-right block sends f_K to $-\kappa f_K$. Thus its exponential sends f_K to $e^{-\kappa z} f_K$, and multiplication by the two off-diagonal blocks gives $-\kappa e^{-\kappa z} T$. The first exponential truncates because $R^2 = 0$. For the second, induction gives $T^j = \kappa^{j-1} T$ for $j \geq 1$; substitution in the entire exponential series proves the formula, including its removable value at $\kappa = 0$.

For direct verification that the memory cancels the extra exponential, substitute $U_R(z) = I_B + zT$ in TDP6 with the blocks in TDP13. The sum of the two terms on its right is $T + z\kappa T - \kappa T \int_0^z e^{-\kappa(z-t)}(1+t\kappa)dt = T$, since the integrand is the derivative in t of $te^{-\kappa(z-t)}$. This is precisely $U'_R(z)$.

Put $c = i \log p$, $n_E = \dim E = q$ and $b_B = \dim B$. The exact support prime action $I_E + cR$ and all its prime powers satisfy

$$\begin{aligned}\Phi((I_E + cR)^m) &= I_B + mcT, \\ (I_B + cT)^m &= I_B + \begin{cases} ((1 + c\kappa)^m - 1)T/\kappa, & \kappa \neq 0, \\ mcT, & \kappa = 0, \end{cases} \\ \det(I_E - w(I_E + cR)) &= (1 - w)^{n_E}. \end{aligned} \tag{TDP14}$$

The first line is the binomial theorem with $R^2 = 0$. For the second line expand by the binomial theorem and use $T^j = \kappa^{j-1}T$; the scalar binomial formula supplies the coefficient. All eigenvalues of R are zero, so all eigenvalues of $I_E + cR$ equal one; this proves the last line. In contrast, the observed block has a candidate eigenvalue $1 + c\kappa$. This is a compression effect, not a moved eigenvalue of the full support action.

When $K \neq 0$, its original block is $I_K + c\epsilon f_K e_K^*$ and has the candidate eigenvalue $1 - c\kappa$. Its exact factors in TDP9 are

$$\begin{aligned}\det(I_B - w(I_B + cT)) &= (1 - w)^{b_B-1}(1 - w - w\kappa), \\ \det(I_K - w(I_K + c\epsilon f_K e_K^*)) &= (1 - w)^{n_E-b_B-1}(1 - w + w\kappa), \\ \mathcal{E}_{p,R}(w) &= \frac{(1 - w)^2}{(1 - w)^2 - w^2 c^2 \kappa^2}. \end{aligned} \tag{TDP15}$$

These identities first hold formally at $w = 0$. The first two follow from the rank-one determinant identity $\det(I + xy^*) = 1 + y^*x$, which is proved by extending x to a basis when $x \neq 0$; the case $x = 0$ is immediate. Their product, the last line and TDP14 multiply to an identity. Alternatively substituting the blocks of TDP13 into TDP9 gives the last line directly. In the inverse determinant, its factor $\mathcal{E}_{p,R}(w)^{-1}$ cancels the two candidate poles at $1/(1 + c\kappa)$ and $1/(1 - c\kappa)$. If $\kappa = 0$ this factor is identically one. If $K = 0$, then $r = 0$, $\kappa = 0$ and the kernel factor is absent; TDP14 remains valid. This is a complete pole-cancellation calculation on the actual support receiver.

The original estimates do not authorize deleting κ :

$$\begin{aligned}|\kappa|^2 &= \epsilon^2 |r|^2 \leq \epsilon^2 \alpha (1 - d), \quad \|T\| = \epsilon \sqrt{ad}, \\ \alpha &\leq e^{-2q\psi(t_N) + O_{h,\varpi}(k \log(q+2))}, \quad \log \epsilon = q\psi(t_N) + O_{h,\varpi}(k \log(q+2)), \\ |\kappa| &\leq e^{O_{h,\varpi}(k \log(q+2))} \sqrt{1 - d}, \quad t_N = (N + 1 - q)/q. \end{aligned} \tag{TDP16}$$

The first inequality is the Cauchy–Schwarz inequality on K using TDP12; the rank-one norm gives the second equality. Substitution of the two original estimates proves the last line. The large rate in ϵ cancels the square-root leakage rate, and the phase of r has not been restricted. Therefore the displayed estimate supplies neither a sign nor a vanishing assertion for κ .

The full $M = C + R$ in ST3 must still be used. For λ outside the spectrum of the original self-adjoint C , rank-one elimination gives

$$\det(\lambda I_E - M) = \det(\lambda I_E - C)[1 - \epsilon e^\dagger(\lambda I_E - C)^{-1}f]. \tag{TDP17}$$

Factor $\lambda I - C$ on the left and use the rank-one determinant identity just proved. Clearing denominators extends the equality to a polynomial identity. Thus nilpotence of R by itself does not remove the spectral displacement in TDP3. In particular products $RC^j R = \epsilon^2(e^\dagger C^j f)f e^\dagger$ need not vanish. The exact interaction with C , not R in isolation, determines the original source spectrum.

B.4 Tensor products retain the same defect, with exact mixed sectors

Let $\nu \geq 1$ be an independent tensor count and retain the product metric $G_N^{\otimes \nu}$. Define

$$\begin{aligned} J_\nu &= J^{\otimes \nu}, \quad P_\nu = P^{\otimes \nu}, \quad Q_\nu = I - P_\nu = \sum_{\emptyset \neq S \subset \{1, \dots, \nu\}} \bigotimes_{j=1}^\nu \begin{cases} Q, & j \in S, \\ P, & j \notin S, \end{cases} \\ M_\nu &= \sum_{j=1}^\nu I^{\otimes(j-1)} \otimes M \otimes I^{\otimes(\nu-j)}, \quad H_\nu = \sum_{j=1}^\nu I^{\otimes(j-1)} \otimes H \otimes I^{\otimes(\nu-j)}, \\ J_\nu^\dagger e^{zM_\nu} J_\nu &= U(z)^{\otimes \nu}, \quad e^{zH_\nu} = (e^{zH})^{\otimes \nu}. \end{aligned} \tag{TDP18}$$

The projections in the first line have mutually orthogonal ranges: two distinct subsets place $PQ = 0$ in at least one factor. Expansion of $(P + Q)^{\otimes \nu}$ proves their sum. The summands of M_ν commute, so multiplication of their convergent exponential series proves the last line. Tensoring therefore preserves the full source action, and also preserves the difference caused by observation:

$$U(z)^{\otimes \nu} - (e^{zH})^{\otimes \nu} = \sum_{j=1}^\nu U(z)^{\otimes(j-1)} \otimes (U(z) - e^{zH}) \otimes (e^{zH})^{\otimes(\nu-j)}. \tag{TDP19}$$

The right side telescopes after inserting and subtracting tensors with successively one further factor $U(z)$. No mixed-kernel subset is discarded by TDP18 or TDP19.

For the original support put $A = \Phi(\Pi) = [u, v][u, v]^*$, with its two strictly positive eigenvalues ϑ_+, ϑ_- from ST7. Let $A_\nu = A^{\otimes \nu}$, which is the exact compression of $\Pi^{\otimes \nu}$. Its eigenvalues and defect are

$$\vartheta_+^{\nu-j} \vartheta_-^j \quad \text{with multiplicity } \binom{\nu}{j}, \quad 0 \leq j \leq \nu, \quad \text{and } 0 \text{ on the remaining } b_B^\nu - 2^\nu \text{ dimensions,}$$

$$A_\nu - A_\nu^2 = \sum_{j=1}^\nu (A^2)^{\otimes(j-1)} \otimes (A - A^2) \otimes A^{\otimes(\nu-j)} = (Q_\nu \Pi^{\otimes \nu} J_\nu)^\dagger (Q_\nu \Pi^{\otimes \nu} J_\nu). \tag{TDP20}$$

Tensor products of orthonormal eigenvectors prove the first line, including labelled multiplicities when values coincide. A telescoping expansion proves the first equality in the second line. Since $\Pi^{\otimes \nu}$ is an orthogonal projection, insertion of $P_\nu + Q_\nu$ proves the last equality. Each summand of the middle expression is positive semidefinite. Positivity of this defect does not make it zero.

There are exact amplification values, not merely an unspecified error:

$$\begin{aligned} \det^+ A_\nu &= \mathfrak{d}^{\nu 2^{\nu-1}}, \quad \frac{\log \det^+ A_\nu}{\nu 2^\nu} = \frac{1}{2} \log \mathfrak{d}, \\ \left| \frac{\log \det^+ A_\nu}{\nu 2^\nu} - \frac{1}{2} \log d \right| &\leq \frac{\alpha}{2\Theta_N}, \quad 0 \leq 1 - \vartheta_+^\nu \leq \nu \alpha, \\ \frac{1}{\nu} \log(\vartheta_+^{\nu-j} \vartheta_-^j) &= (1 - j/\nu) \log \vartheta_+ + (j/\nu) \log \vartheta_-. \end{aligned} \tag{TDP21}$$

Each of ϑ_+ and ϑ_- occurs $\nu 2^{\nu-1}$ times in the product of the 2^ν nonzero eigenvalues, proving the determinant identity. ST11 then proves its estimate. The identity $1 - x^\nu = (1 - x) \sum_{j=0}^{\nu-1} x^j$ for $0 \leq x \leq 1$, together with ST8, proves the bound on the first tensor branch. The final line is the logarithm of the exact eigenvalue. In particular a non-unit second support eigenvalue contributes a loss proportional to the tensor count. Taking a ν th root does not remove that value. This is the precise contrast with a fixed exponent error divided by the tensor count in Deligne's amplification.

The support representation survives observation as a homomorphism exactly when both original support vectors are retained:

$$\Phi \circ \rho \text{ is multiplicative} \iff a = d = 1 \iff e, f \in K^\perp. \tag{TDP22}$$

For necessity, multiplicativity makes $A^2 = A$. Its two nonzero eigenvalues must then be one. Thus $a + d = \text{tr } A = 2$ and $0 \leq a, d \leq 1$ imply $a = d = 1$; Cauchy–Schwarz on K gives $r = 0$. The norm identities in TDP12 imply $e, f \in K^\perp$. Conversely, in that case both Π and R preserve $K^\perp$ and are zero on K . Their compressions obey the identical multiplication table $A^2 = A$, $AT = TA = T$, $T^2 = 0$. The complement $I_B - A$ has zero products with A, T , so the algebra formula in TDP11 is multiplicative. The defect space determined by failure is the actual range $\text{im}(Q\Pi J) \subset K$, with Gram operator $A - A^2$, and its tensor analogues are the explicitly decomposed ranges in TDP20.

The support nilpotent also tensors without dropping mixed terms. Put $R_\nu = \sum_{j=1}^\nu I^{\otimes(j-1)} \otimes R \otimes I^{\otimes(\nu-j)}$. Commutativity between different factors and $R^2 = 0$ give

$$R_\nu^j = j! \sum_{|S|=j} \bigotimes_{l=1}^\nu \begin{cases} R, & l \in S, \\ I_E, & l \notin S, \end{cases} \quad (0 \leq j \leq \nu), \quad R_\nu^{\nu+1} = 0, \quad R_\nu^\nu = \nu! R^{\otimes \nu} \neq 0, \quad (\text{TDP23})$$

$$e^{zR_\nu} = (I_E + zR)^{\otimes \nu}, \quad \Phi_\nu(e^{zR_\nu}) = (I_B + zT)^{\otimes \nu}, \quad \Phi_\nu(X) = J_\nu^\dagger X J_\nu.$$

In expanding the j th power, only words using j distinct positions survive; every subset occurs in $j!$ orders. Since $R \neq 0$, choose x with $Rx \neq 0$; then $R^{\otimes \nu} x^{\otimes \nu} \neq 0$, proving its nonvanishing. Thus the full support tensor has only eigenvalue one after exponentiation, while its nilpotent length is exactly $\nu + 1$. Observation keeps the explicitly displayed polynomial in z of degree at most ν . It cannot be replaced by $e^{z\Phi_\nu(R_\nu)}$ without the memory already calculated.

B.5 The exact object carrying eigenvalues through observation

For each $\lambda \in \mathbb{C}$, define the lifted observed eigenspace and its observed image by

$$\begin{aligned} \mathcal{L}_\lambda &= \{(y, z) \in B \oplus K : (H - \lambda I)y + B_{01}z = 0, \quad B_{10}y + (D_K - \lambda I)z = 0\}, \\ \mathcal{O}_\lambda &= \{y \in B : \text{there is } z \in K \text{ with } (y, z) \in \mathcal{L}_\lambda\}, \\ 0 &\longrightarrow \ker(D_K - \lambda I) \cap \ker B_{01} \xrightarrow{z \mapsto (0, z)} \mathcal{L}_\lambda \xrightarrow{(y, z) \mapsto y} \mathcal{O}_\lambda \longrightarrow 0, \\ \mathcal{L}_\lambda &\xrightarrow[\cong]{W} \ker(M - \lambda I), \quad \|W(y, z)\|_{G_N}^2 = \|y\|^2 + \|z\|^2. \end{aligned} \quad (\text{TDP24})$$

Multiplication by the block matrix in TDP5 proves the last isomorphism. The first map in the exact sequence is injective; the kernel of the second consists exactly of pairs $(0, z)$ satisfying both displayed equations; and surjectivity is the definition of $\mathcal{O}_\lambda$. Orthogonality in TDP4 proves the metric identity. If $\lambda \notin \text{spec } D_K$, this same space is the explicit graph

$$\begin{aligned} z &= (\lambda I - D_K)^{-1} B_{10}y, \quad \mathcal{O}_\lambda = \ker[H + B_{01}(\lambda I - D_K)^{-1} B_{10} - \lambda I_B], \\ \|(y, z)\|^2 &= y^* [I_B + B_{10}^\dagger (\bar{\lambda} I - D_K^\dagger)^{-1} (\lambda I - D_K)^{-1} B_{10}] y. \end{aligned} \quad (\text{TDP25})$$

Solving the second equation of TDP24 proves the first line, and taking the squared norm proves the second. At exceptional λ TDP24 remains the exact space, with no discarded singular fibre.

For a nonzero original eigenvector x , set $y = J^\dagger x$ and $z = \iota_K^\dagger x$. Even when the programme proves $y \neq 0$ for its terminal class, the exact observed equation is $(H - \lambda I)y = -B_{01}z$, not an eigenvalue equation for H alone. Tensor products preserve the lift: $x^{\otimes \nu}$ maps to $y^{\otimes \nu}$ in the fully observed sector, and to every other subset sector of TDP18 with the appropriate y or z factor. If $y \neq 0$, its exact detection ratio is

$$\frac{\|J_\nu^\dagger x^{\otimes \nu}\|}{\|x^{\otimes \nu}\|} = \left(\frac{\|y\|}{\sqrt{\|y\|^2 + \|z\|^2}} \right)^\nu. \quad (\text{TDP26})$$

This follows from the definition of the Hilbert tensor metric and TDP24. A fixed non-unit detection factor thus gives a linear logarithmic cost in the tensor count; it is not a fixed additive error that disappears after taking a tensor root. If $y = 0$, the full lift still exists but its fully observed tensor sector is zero. No lower bound for that case has been invented.

B.6 What this changes in a Deligne-style amplification

Deligne's two relevant mechanisms have different exact inputs. In 1.4.3–1.5.2 the cohomological denominator has a bounded weight shift; the trace formula identifies it with the Euler product; nonnegative coefficients for the even tensor powers prevent a local pole from being hidden inside the product. The finite tensor calculation $w(\alpha^{2\nu}) = 2\nu w(\alpha)$ then divides the fixed error by 2ν . In the supplied version of 3.2, an eigenvalue on H_c^1 squares inside a Künneth component of a surface, while the surface estimate improves the induction error. Its displayed inequalities have the arithmetic implication

$$2w(\alpha) < 2 + 2^{-j} \implies w(\alpha) < 1 + 2^{-j-1}. \quad (\text{TDP27})$$

This line is just division by two. Attribution of the full surrounding proof remains to Deligne; the local abbreviated English source is not used to supply omitted hypotheses or maps.

For the present programme, TDP3 gives the actual prime eigenvalues, TDP24 gives their exact observation map, TDP18 gives the full tensor object, and TDP9 gives the exact local determinant after observation. The kernel contribution is the concrete rational function $\mathcal{E}_p$, not an unnamed missing term. Its support-only value in TDP15 cancels precisely the false poles that discarding the kernel would introduce. Accordingly the positivity of $A - A^2$ in ST9 cannot replace the noncancellation input in Deligne's pole argument: TDP15 is an exact example with a positive support defect and nontrivial pole cancellation. Likewise TDP21 and TDP26 calculate the tensor loss itself; the second support eigenvalue and the eigenvector's unobserved component are retained as spaces and maps, not removed by a weight declaration.

All formulas apply separately at $N = q - 1, q, 2q - 1, 2q$. Any signed return uses the original signs $+, +, -, -$; for scalar absolute error estimates the triangle inequality gives the sum of the four respective errors. No cancellation between these cutoffs has been assumed. No equivalence between the original conductor and Deligne's full cohomological trace formula, and no conclusion about RH, is claimed from a compressed positivity assertion. The calculation instead supplies the exact prime-power, determinant and tensor defect objects that the original programme now has to evaluate, with their connecting maps proved.

There is an exact positive receiver for every finite original prime combination, with the adjoint in the original metric. Let $\mathcal{I}$ be any finite set of prime-power indices (p, m) , let $c_{p,m} \in \mathbb{C}$ be arbitrary, and put $X = \sum_{(p,m) \in \mathcal{I}} c_{p,m} F_p^m$. Then

$$\begin{aligned} \Phi(X^\dagger X) - \Phi(X)^\dagger \Phi(X) &= (QXJ)^\dagger (QXJ) \\ &= \sum_{(p,m),(r,n) \in \mathcal{I}} \overline{c_{p,m}} c_{r,n} J^\dagger (F_p^m)^\dagger Q F_r^n J \succeq 0. \end{aligned} \quad (\text{TDP28})$$

Insertion of $I_E = P + Q$ between $X^\dagger$ and X proves the first identity, since $Q^2 = Q = Q^\dagger$ and $P = JJ^\dagger$. Expansion proves the second identity, with all phases retained. Evaluation on $b \in B$ gives $\|QXJb\|_{G_N}^2$, proving positivity. This is an exact prime-action Gram identity. Equating it to Weil's scalar positivity functional would require an additional proved map; no such equality is asserted here. The positive adjoint-product defect and the Euler product defect in TDP9 are both calculated, and they have different factors: the latter contains $B_p(I - uD_p)^{-1}C_p$, without an adjoint.

C Source identities and proof locations

1. *A Note on the Globalization Semiring $G(\mathbb{Z})$* , March 2026, exact original source with empty author field, [pinned complete original TeX](#). The source is read in full and retained unchanged. Corollaries `cor:prime-ideals`, `cor:dim-two`, and Theorem `thm:zero-prime-not-irr` already contain the scalar prime results. The user's Gemini 2.5 Pro origin attribution is recorded above. The distinct deposit cited in Version 11 is [DOI 10.5281/zenodo.18976982](#); its

metadata is not substituted for the supplied file’s author field. The class-semigroup assertion of that original note is not used here: its displayed scalar set includes the supported ring zero. Indeed for every supported ideal I_J , multiplication by that zero gives $0I_J = \{0, \tau\}$; choosing both allowed scalars equal to zero identifies every pair of supported ideals in its stated equivalence. The later Version 11 normed ideal-class definition is a different, restricted definition.

2. The Clankers, *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, pre-publication Version 11, June 2026. Complete supplied source: `sources/FULL_LATTICE_FOUNDATION_V11.tex`; exact original proof labels and source reading coverage are retained in the source ledger. The spectrum and branch theorems are used with their existing proofs, not represented as new.
3. Garrett Birkhoff, *Lattice Theory*, third edition, American Mathematical Society, 1967. Lattice-theoretic background credited by Version 11. The finite and infinite separation arguments needed here are written out in FLS7 and FLS13–18.
4. Programme result SZ-20260922-028, *What the original observation retains of split support, complete ST1–20 proof, pinned edition*. Root downloaded this exact public file and verified equality with the retained source bytes. The original objects and metrics are ST1–3; the scalar action is ST4; the finite bounds are ST6–20. Their human sources are cited in that proof. FSJ1–13 derives the full-lattice extension of its scalar action.
5. Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, [arXiv:math/9811068v1](https://arxiv.org/abs/math/9811068v1), Section VI, equations (1)–(16). Original downloaded author TeX and unchanged source archive retained. Used in FLC1–5.
6. Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022v1](https://arxiv.org/abs/2602.04022v1), section `sectweilpos`, equations `wp` and `arch`, and its statement of André Weil’s criterion. Original author TeX read at these locations; used in FLC14–21, PDD1–15 and GPE5–12.
7. Pierre Deligne, *La conjecture de Weil. II*, Publications Mathématiques de l’IHÉS **52** (1980), 137–252, [original publication record](#). The local supplied English TeX is a transcription/translation, not author LaTeX. Only the reading range stated in the TDP appendix is claimed; the new finite-dimensional identities in that appendix are proved there in full.